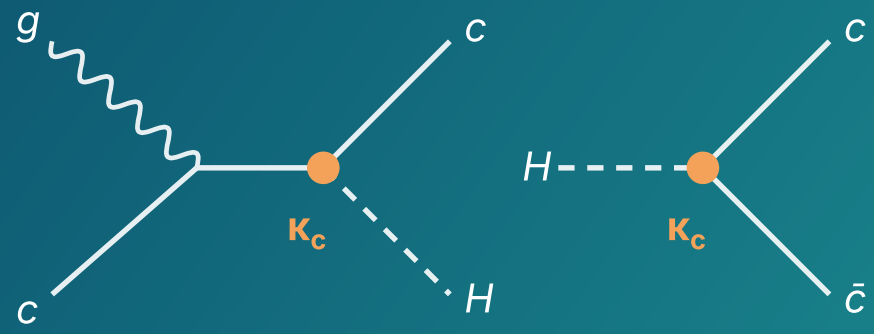
# Constraints on Higgs–charm couplings at CMS

*Higgs 2026 · Louvain-la-Neuve · 28 October 2026*

**Chirayu Gupta**
*Vrije Universiteit Brussel*

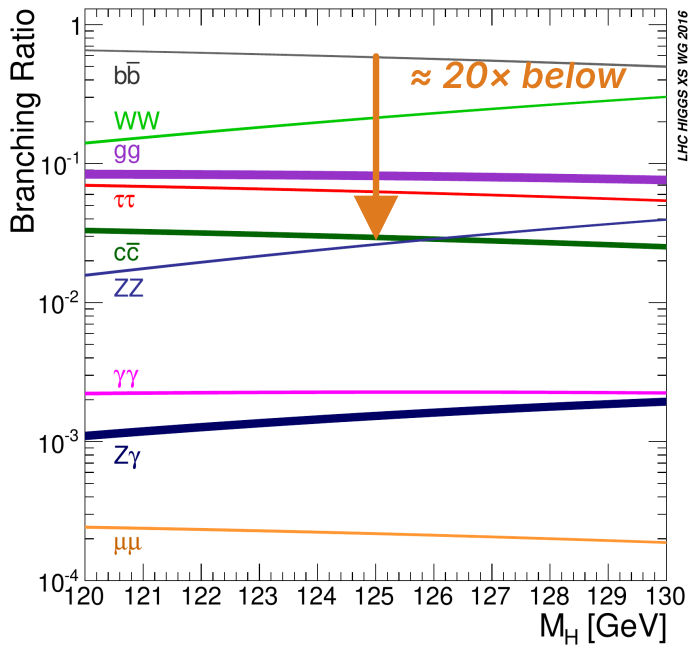
on behalf of the CMS Collaboration



# Why the Higgs–charm coupling?

## Second-generation quark Yukawas are the open frontier

- Since 2012: couplings to **t, b, τ** measured, all SM-like
- Evidence for **H → μμ** (backup)
- **CHARM IS THE MOST ACCESSIBLE SECOND-GENERATION QUARK**




SM branching ratios: arXiv:1610.07922

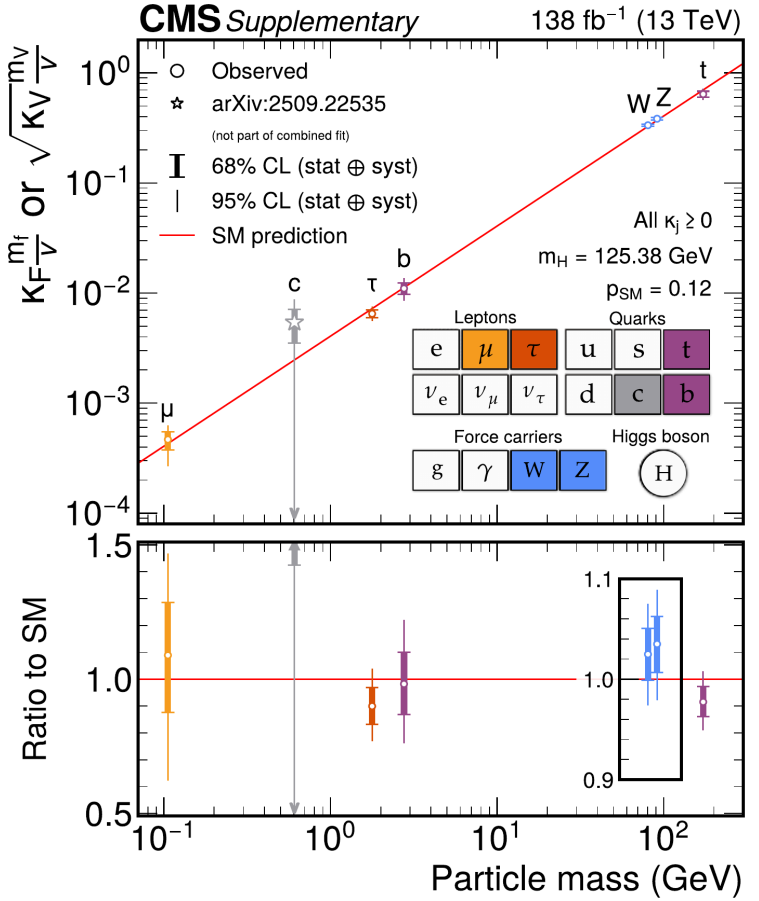
$\mathcal{B}(H \to c\bar{c}) \approx$ **2.9%**, about **20× below** $b\bar{b}$, with huge QCD backgrounds and hard c tagging (slide 3)

$$\kappa_c = y_c / y_c^{SM} \quad \text{(SM: 1)}$$

Flavour-compatible BSM, e.g. spontaneous flavour violation, allows $|\kappa_c| \sim 3$:

**CURRENT SENSITIVITY IS ENTERING THIS REGION**

SFV: arXiv:1811.00017
$\kappa_c$ in SFV: arXiv:2410.05236




CMS, arXiv:2602.18611 (suppl.); $\kappa_c$ point: separate $t\bar{t}H$+VH result, not in the fit

**➜ *Does the Higgs give charm its mass?***

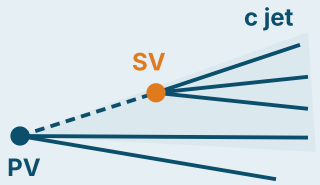
CMS-DP-2025-081 · arXiv:2202.03772

# Charm tagging

## Finding c jets between b and light jets

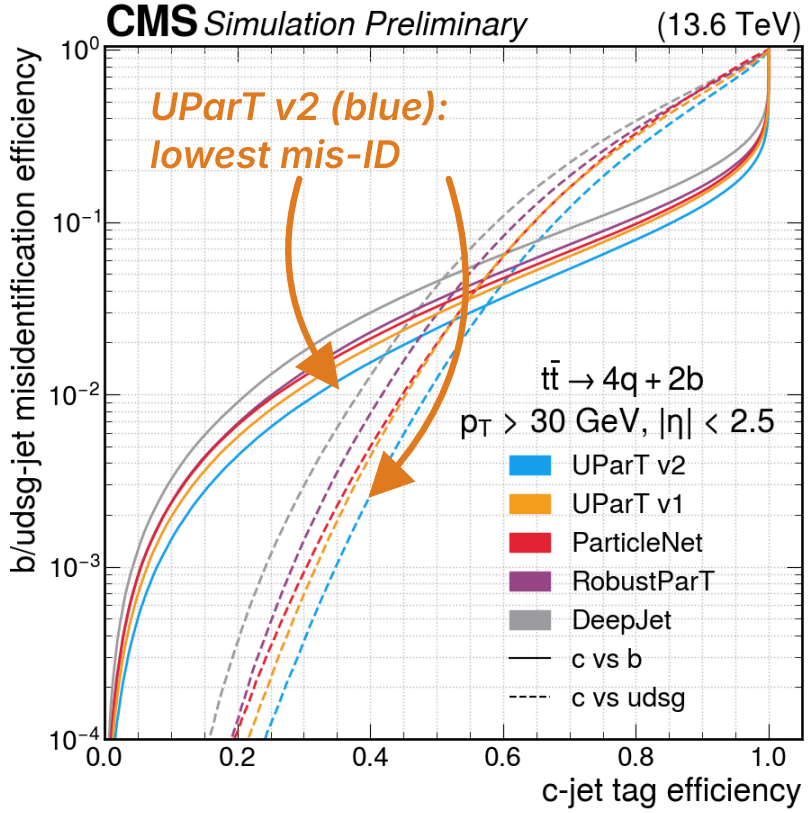
### Charm signature

c-hadron mass and lifetime lie **between light and b**: a secondary vertex (SV) like b jets, but closer to the primary vertex, so **easily mistaken for light jets**




### State of the art in Run 3: UParT

- **Transformer** on the jet's particles, with pairwise particle-interaction features (ParT architecture)
- **UParT v2**: lowest b and light mis-identification at fixed c-jet efficiency (right)
- The Run 2 results in this talk use the previous generation (below)




$t\bar{t}$ simulation, 13.6 TeV · solid: c vs b, dashed: c vs light

**In this talk** DeepJet · VH resolved, cH | ParticleNet · VH merged, $t\bar{t}H$ | DeepDoubleX · boosted H

# How CMS probes $\kappa_c$

## In the decay, in production, and through the Higgs width

### Direct H → $c\bar{c}$

slides 5–11

rate ∝ $\kappa_c^2/\Gamma_H$ · VH, $t\bar{t}H$, boosted (mostly ggH)

**$|\kappa_c| < 3.5$ (2.7)** $t\bar{t}H + VH$

arXiv:2205.05550 · arXiv:2509.22535 · arXiv:2211.14181

### H + charm production

slides 12–14

cg → cH: $\kappa_c$ in the production vertex · H → γγ, WW

**$|\kappa_c| < 47$ (51)**

arXiv:2503.08797 · arXiv:2508.14988

### Indirect: Higgs rates and width

slide 15

on-shell and NEW off-shell H → ZZ → 4ℓ; range depends on assumptions

**$\kappa_c \in [-3.3, 2.8] \ldots [-8.6, 7.5]$**

arXiv:2502.05665 · arXiv:2607.23352

### Rare exclusive decays

slide 16

H → J/ψ γ, ψ(2S) γ: sensitive to the *sign* of $\kappa_c$

**$-166 < \kappa_c < 208$**

arXiv:2411.15000

NEW = released in 2026 · μ = $\sigma\mathcal{B}/(\sigma\mathcal{B})_{SM}$ · $\Gamma_H$ = total Higgs width · μ limits and $\kappa_c$ intervals at 95% CL, observed (expected)

***→ All $\kappa_c$ results: Run 2 (13 TeV, up to 138 $fb^{-1}$) · Run 3 so far: better triggers (slide 17)***

# VH(H → $c\bar{c}$): strategy

arXiv:2205.05550

## Cleanest signature: the W or Z decays to leptons

**0ℓ · Z → νν** **1ℓ · W → ℓν** **2ℓ · Z → ℓℓ**

Main backgrounds V+jets and $t\bar{t}$, normalised in control regions

*below* **$p_T(H_{cand})$ = 300 GeV** *above*

### Resolved · two small-R jets (R = 0.4)

- **>95% of the signal**, but more background
- **DeepJet** CvsL, CvsB: the two highest-CvsL jets form $H_{cand}$
- Mass peak up to **20% sharper**: c-jet energy regression, FSR-jet recovery, kinematic fit (2ℓ)

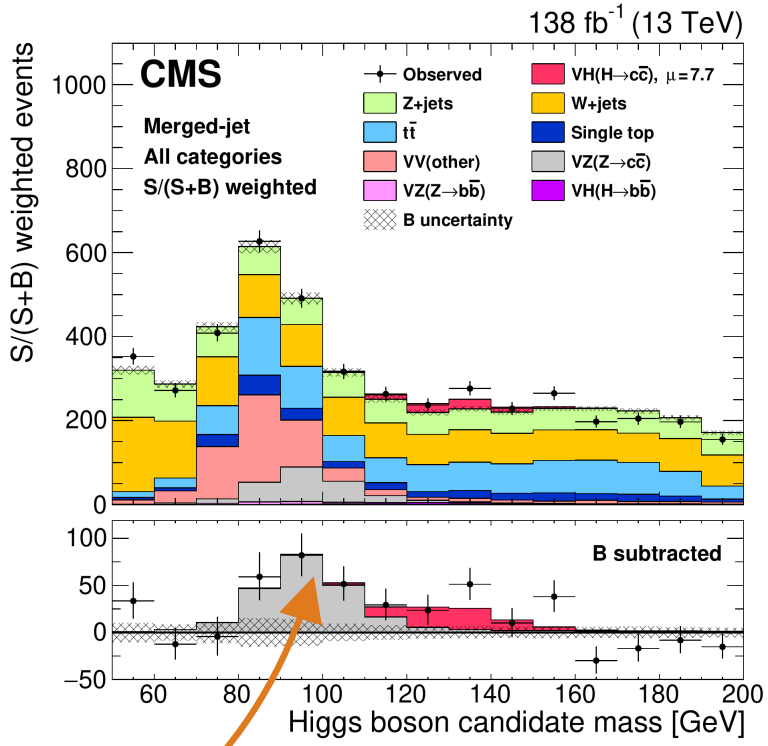
**Fit: BDT score (m($H_{cand}$) among the inputs)**

### Merged · one large-R jet (R = 1.5)

- **<5% of the signal**, but much less background
- **ParticleNet** $c\bar{c}$ tagger (graph NN, mass-decorrelated): **4–7×** better rejection than DeepAK15
- ParticleNet mass regression: **≈50% better** mass resolution

**Fit: m($H_{cand}$) (BDTs decorrelated from it)**

Merged-jet mass; bottom: VZ (grey, ≈ 90 GeV) and VH (red) after background subtraction

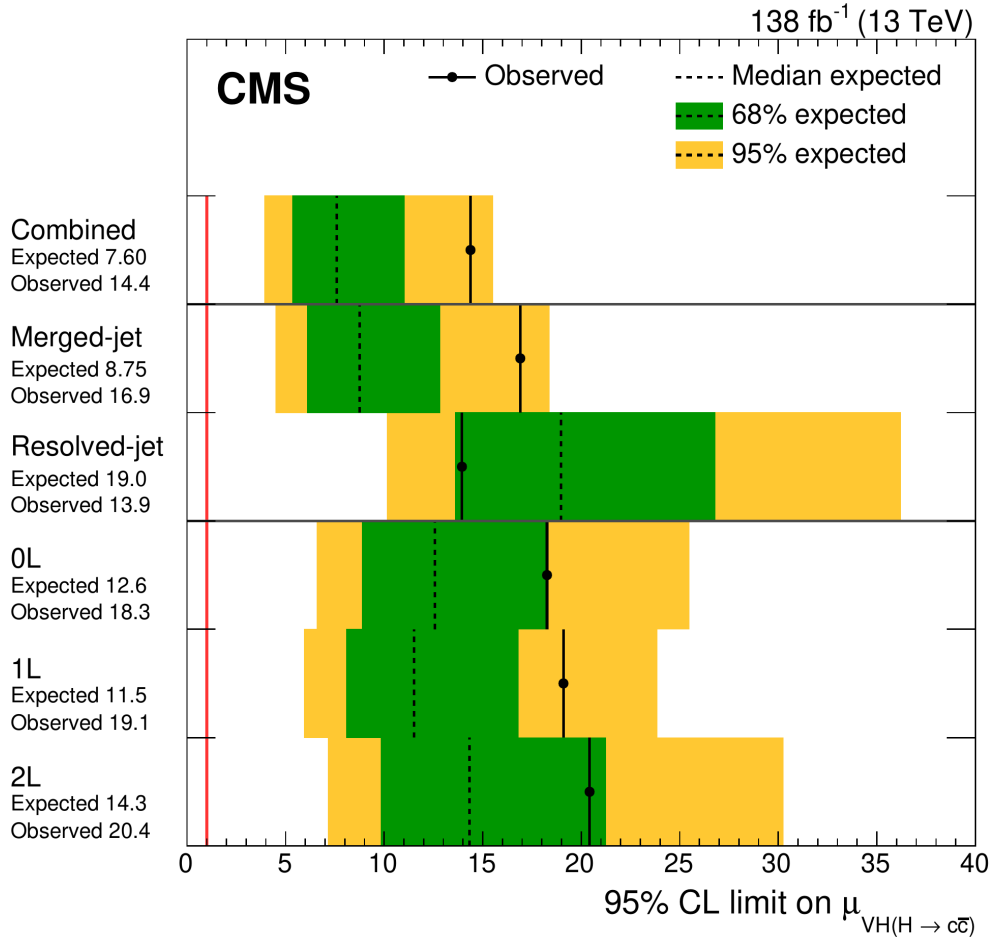



**Validation:** the same analysis sees **VZ(Z → $c\bar{c}$) at 5.7σ** (grey peak, right): c tagging and background model work → backup

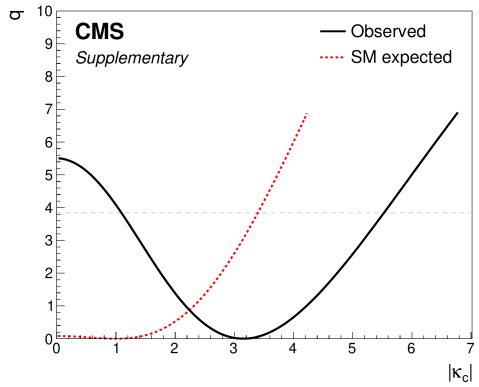
# VH(H → $c\bar{c}$): results

arXiv:2205.05550

## Best expected μ of any single Run 2 channel




95% CL limits on μ per category · red line: SM (μ = 1) · the **merged-jet** category carries most of the sensitivity




q = −2Δln L vs $|\kappa_c|$ (supplementary) · 95% CL where q < 3.84 (dashed line)

95% CL · observed (expected)

**μ < 14** (7.6) **1.1 < $|\kappa_c|$ < 5.5** (< 3.4)

μ limits with CLs; $\kappa_c$ interval from the profile-likelihood scan

**0.85**

**$\Delta\mu_{stat}$ / $\Delta\mu_{total}$**

**Remove every systematic uncertainty: Δμ shrinks by only 15%**

| | |
|---|---|
| Statistical | 0.85 |
| Experimental | 0.48 |
| Theory | 0.22 |

$\Delta\mu/\Delta\mu_{total}$ per source (Table 1), in quadrature

***➜ Expected μ < 7.6: the best of any single Run 2 channel***

# Boosted H → $c\bar{c}$

arXiv:2211.14181

## Gluon-fusion enriched: one large-radius jet at high $p_T$

### ① Boosted regime: $p_T^H$ > 450 GeV

- High $p_T$ suppresses the QCD background
- H → $c\bar{c}$ in **one large-radius jet** (R = 0.8); soft drop grooms it and sharpens the mass $m_{SD}$
- Signal: 55% ggF, 25% VBF, 20% VH

### ② DeepDoubleX tagger

- Convolutional + recurrent network on particle-flow candidates and secondary vertices
- **Passing** region (signal-rich) and **failing** region (background-rich)
- Mass-decorrelated: QCD in the passing region is **predicted from data** in the failing region

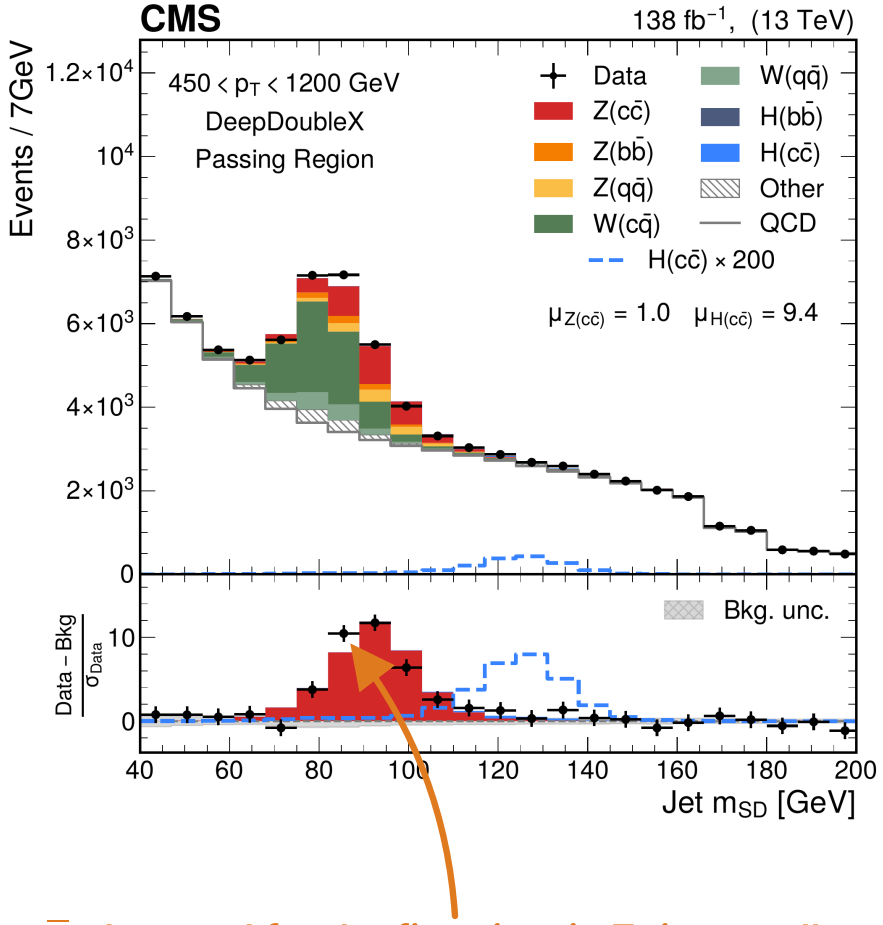
### ③ Statistical analysis

- Simultaneous binned fit of $m_{SD}$ in both regions, in 6 $p_T$ bins
- Upper limit from the profile likelihood ratio (CLs)

**$\sigma\mathcal{B} / (\sigma\mathcal{B})_{SM}$ < 47 (39)**

95% CL · observed (expected) · no $\kappa_c$ interpretation in the paper

Jet mass in the passing region · H → $c\bar{c}$ dashed (×200)




*Z → $c\bar{c}$ observed for the first time in Z+jets, well over 5σ*

# $t\bar{t}H(H \to c\bar{c})$: selection and jet tagging

arXiv:2509.22535

## Three $t\bar{t}$ decay channels, every jet flavour-tagged

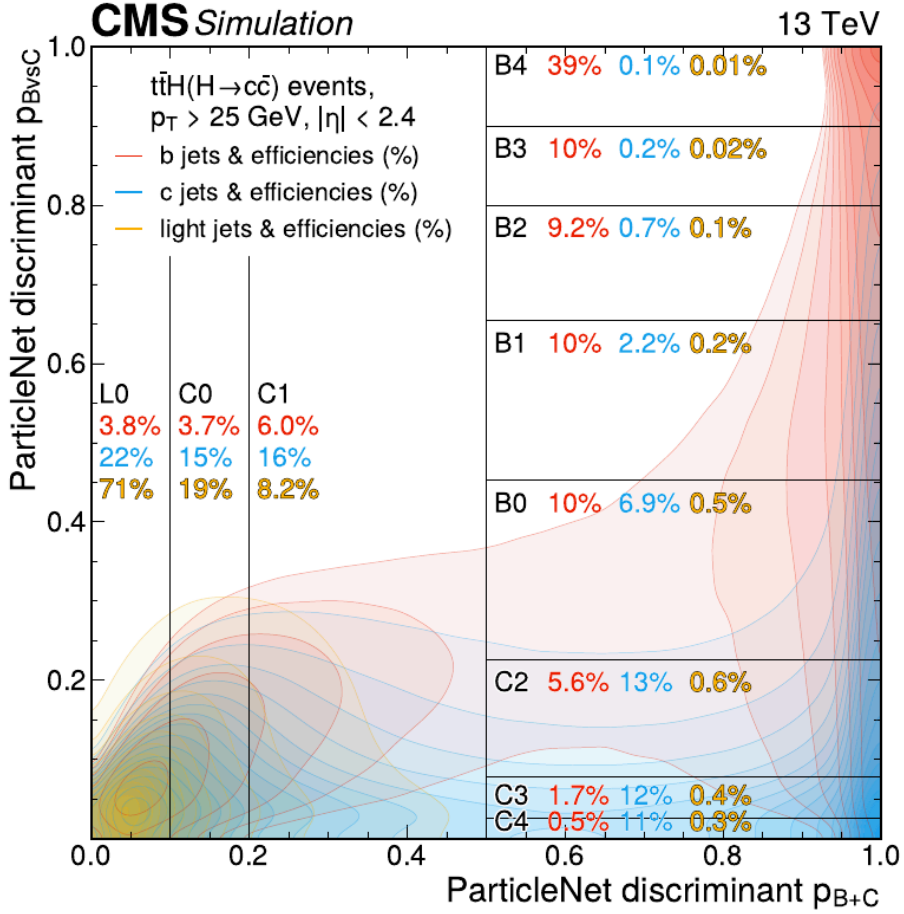
**0ℓ** — **≥ 7 jets**, $H_T$ > 500 GeV
fully hadronic $t\bar{t}$

**1ℓ** — **1 lepton + ≥ 5 jets**
single-lepton $t\bar{t}$

**2ℓ** — **2 opposite-charge leptons + ≥ 4 jets**, Z veto
dilepton $t\bar{t}$

**ParticleNet** (graph NN) tags every jet as b, c or light: **11 categories** (right), up to **2×** better background rejection than DeepJet

**Main background:** $t\bar{t}$+jets, split by the flavour of the extra jets: $t\bar{t}$+b, +≥2b, +c, +≥2c, +light




b, c and light jets in the ParticleNet (b+c vs light, b vs c) plane, with the 11 tagging categories

# $t\bar{t}H(H \to c\bar{c})$: event classifier and regions

arXiv:2509.22535

## Signal, control and validation regions from one classifier

**Event classifier: ParT** (transformer) on jets, leptons and $p_T^{miss}$: one score per process (4 signals, 5 $t\bar{t}$+jets, + QCD in 0ℓ)

$D_{t\bar{t}X}$ = sum of the four signal scores, sorts every event into a region:

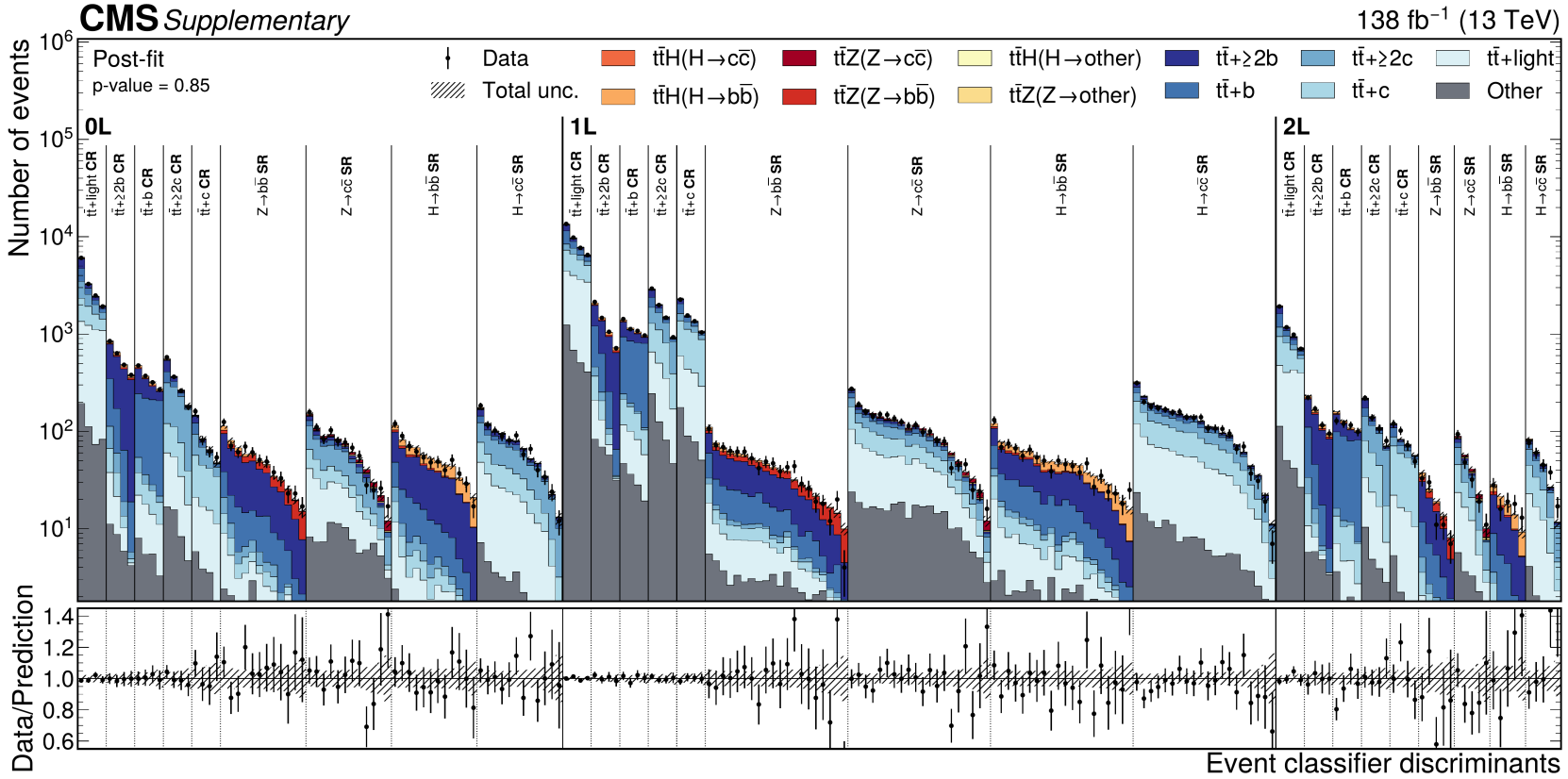
| rejected<br>D < 0.4 | Validation<br>0.4–0.6 | 5 control regions<br>0.6–0.85 | 4 signal regions<br>D > 0.85 |
|---|---|---|---|
| least signal-like events | **signal-depleted**, same SR/CR structure: good agreement → backup | one per $t\bar{t}$+jets flavour: **normalisations float** in the fit | $t\bar{t}H$ / $t\bar{t}Z$ × $c\bar{c}$ / $b\bar{b}$; c-enriched SRs need **≥ 2 c jets** |

- **QCD multijet** (0ℓ): a cut on the QCD score leaves < 1% of $t\bar{t}$+jets
- **Fit**: ParT score in all regions of the three channels at once

# $t\bar{t}H(H \to c\bar{c})$: results

arXiv:2509.22535

## One fit to all regions: four signals and every $t\bar{t}$+jets component




Post-fit ParT discriminants in the 5 CRs and 4 SRs of each channel (0ℓ, 1ℓ, 2ℓ) · fit p-value 0.85

$t\bar{t}H(H \to c\bar{c})$ · 95% CL · observed (expected)

**$\mu < 7.8$ (8.7)**

**$|\kappa_c| < 3.0$ (3.3)**

$\kappa_b = 1$ · with $\kappa_b$ fitted: $|\kappa_c| < 4.2$ obs. → backup

| | best-fit μ | obs. (exp.) |
|---|---|---|
| $t\bar{t}H(b\bar{b})$ | $0.91\ ^{+0.26}_{-0.22}$ | 4.4σ (4.5σ) |
| **$t\bar{t}H(c\bar{c})$** | **−1.6 ± 4.5** | – |
| $t\bar{t}Z(b\bar{b})$ | $1.47\ ^{+0.45}_{-0.41}$ | 3.5σ (2.4σ) |
| $t\bar{t}Z(c\bar{c})$ | $1.02\ ^{+0.79}_{-0.84}$ | 1.2σ (1.3σ) |

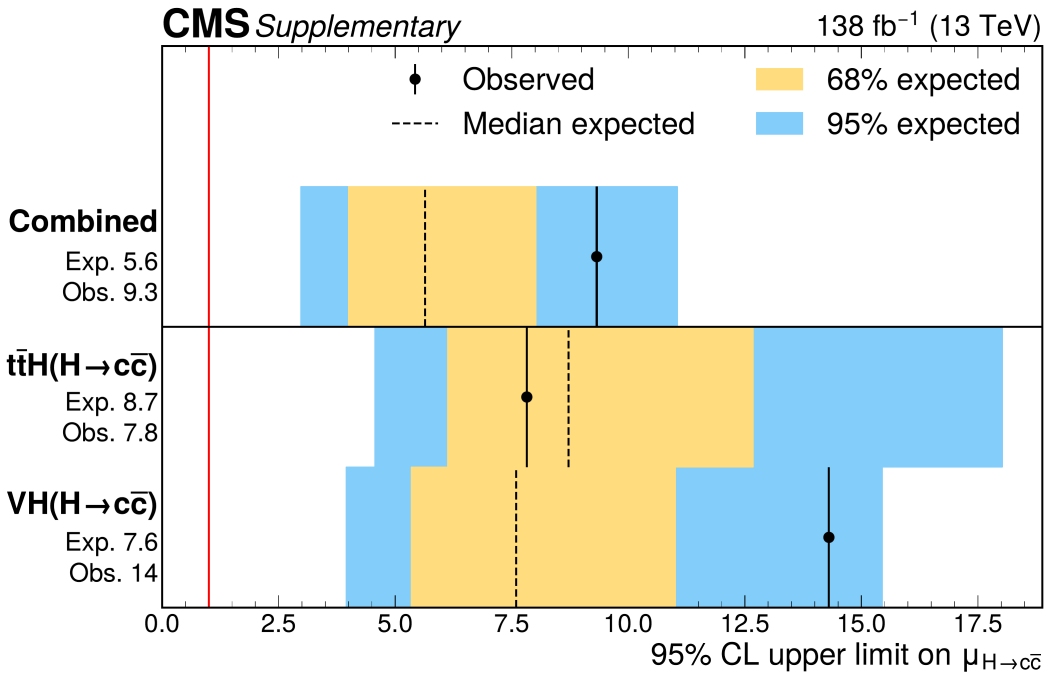
$t\bar{t}Z(c\bar{c})$ and $t\bar{t}Z(b\bar{b})$ are the standard candles: both consistent with the SM · limits per channel and event yields → backup

***→ A second direct channel, with expected sensitivity comparable to VH***

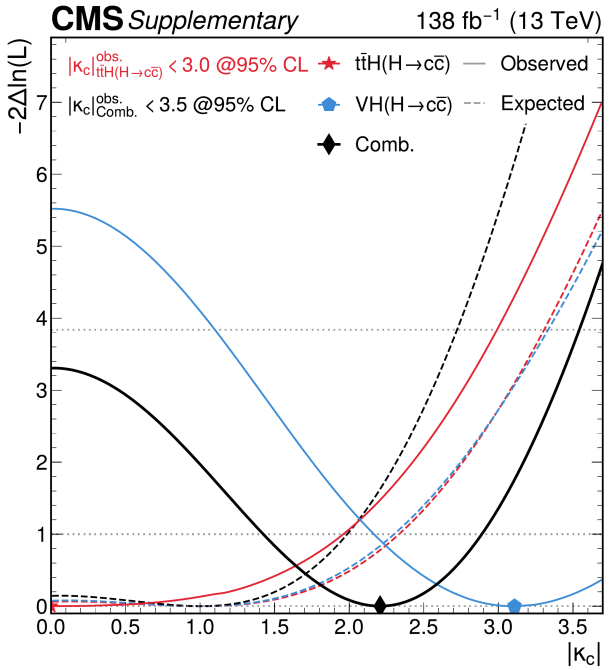
# $t\bar{t}H$ + VH combination

arXiv:2509.22535

## Best expected direct sensitivity to $\kappa_c$




95% CL limits on $\mu_{H\to c\bar{c}}$: $t\bar{t}H$, VH and combined




$-2\Delta\ln L$ vs $|\kappa_c|$ ($\kappa_b = 1$) · red $t\bar{t}H$, blue VH, black comb.

$t\bar{t}H$ + VH · 95% CL
observed (expected)

**$\mu < 9.3$ (5.6)**

**$|\kappa_c| < 3.5$ (2.7)**

SM production, $\kappa_b = 1$

**26%**
better expected μ limit than VH alone

**36%**
better expected μ limit than $t\bar{t}H$ alone

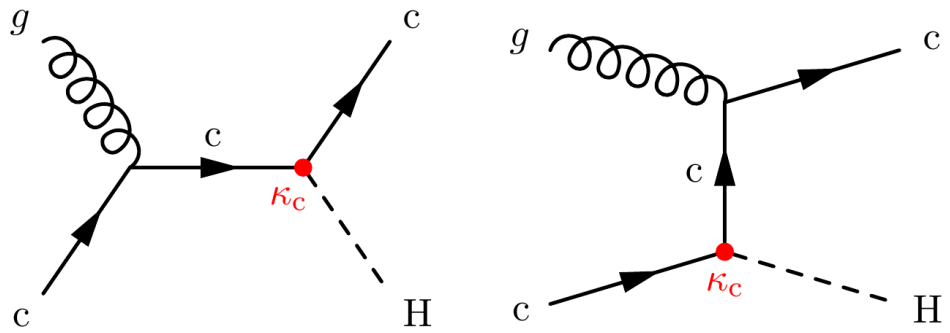
**$|\kappa_c| < 2.7$**
expected, below the BSM-motivated ~3

***➜ The two direct channels together set the best expected constraint on $\kappa_c$***

# H + charm production

arXiv:2503.08797 · arXiv:2508.14988

## $\kappa_c$ in the production vertex, independent of the H → $c\bar{c}$ decay




LO diagrams for cH production; $\kappa_c$ at the marked vertices

**Signal**: a clean Higgs decay (γγ or WW) + one c-tagged jet

## Challenges

### 1 Tiny rate

$\sigma_{cH} \lesssim 100$ fb, ~0.2% of all Higgs production: $\sigma\mathcal{B} \approx$ **0.2 fb** (γγ), ≈ 20 fb (WW)

### 2 Yukawa-insensitive processes

ggH and other Higgs processes with an extra c jet (e.g. from g → $c\bar{c}$) give the **same final state** without involving $\kappa_c$

### 3 b-associated Higgs (bH)

About **7× larger** than cH: b jets mistagged as c, so c-vs-b separation matters

### 4 Modelling

Typical flavour-scheme uncertainty (4FS vs 3FS for cH): **25–30%**; heavy-flavour modelling of ggH (T. Bevilacqua, next talk)

***→ A complementary handle on $\kappa_c$, but small rates and irreducible H+c make it hard***

# cH(H → γγ)

arXiv:2503.08797

## First search for H + charm in the diphoton channel, 138 $fb^{-1}$

### ① Selection
- Two identified photons
- Leading jet ($p_T$ > 25 GeV) **c-tagged** with DeepJet

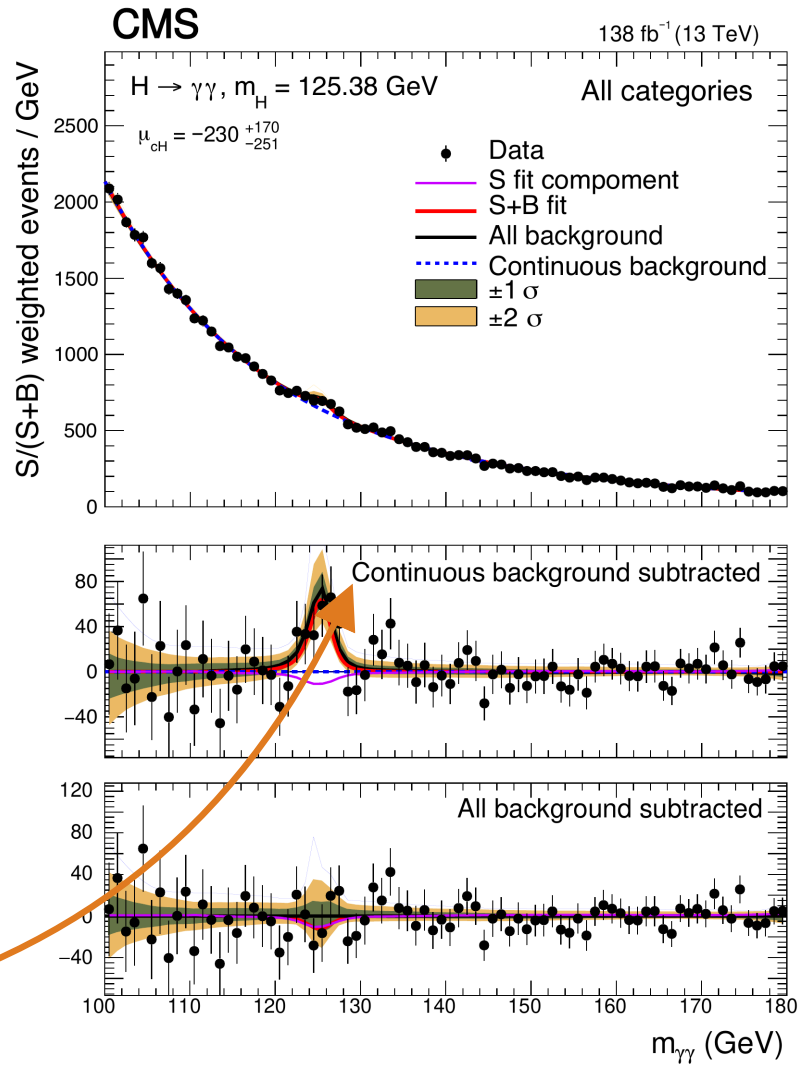
### ② Two BDTs, no c-tag inputs
- cH vs **ggH**, the largest resonant background
- cH vs the **continuum** γγ + jets

### ③ Categories and fit
- 27 categories (9 per year)
- Signal from fits to $m_{\gamma\gamma}$

**$\mu_{cH}$ < 243** (355) **$|\kappa_c|$ < 38** (73)

95% CL · observed (expected)

*The 125 GeV peak is explained by the other Higgs processes (mostly ggH)*




*→ $|\kappa_c| \lesssim 40$: $\sigma\mathcal{B} \approx 0.2$ fb, and Yukawa-insensitive H+c looks the same*

# cH(H → WW) and the γγ + WW combination

arXiv:2508.14988

## eνμν + a c jet, then combined with γγ

### ① Selection

- One electron + one muon, $p_T^{miss}$ > 20 GeV
- ≥ 1 **c jet** (DeepJet): 43% c efficiency, 27% b and 5% light mistag, to separate cH from bH

### ② Two BDTs

- vs non-Higgs backgrounds
- vs other Higgs processes; signal regions with 1 or > 1 c jets

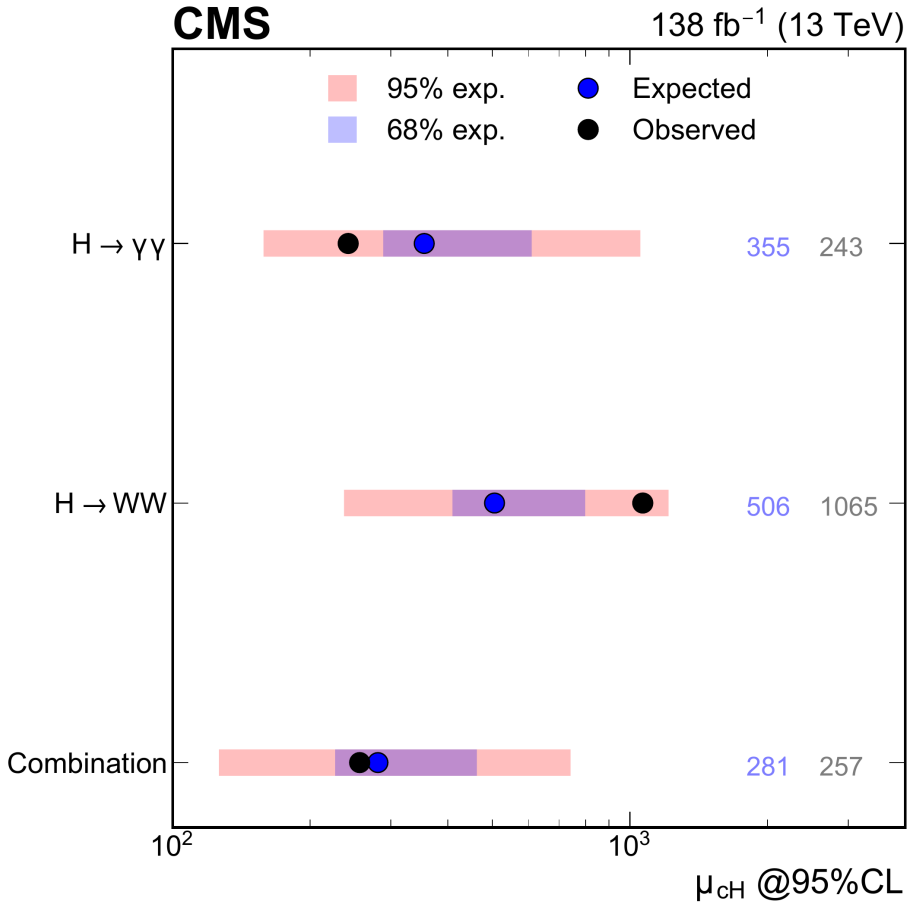
### ③ Fit

- $\mu_{cH}$ is the only parameter; all other Higgs processes fixed to their SM prediction (floating the H+c part → backup)

γγ + WW · 95% CL · observed (expected)

**$\mu_{cH}$ < 257 (281)  $|\kappa_c|$ < 47 (51)**

WW alone: $\mu_{cH}$ < 1065 (506), $|\kappa_c|$ < 211 (95)




$\mu_{cH}$ limits: γγ, WW and combined

***→ $\kappa_c$ from production alone, independent of the H → $c\bar{c}$ decay; statistics-limited***

# Indirect constraints on $\kappa_c$

arXiv:2502.05665 · arXiv:2607.23352

## The Higgs width from on- and off-shell H → ZZ → 4ℓ

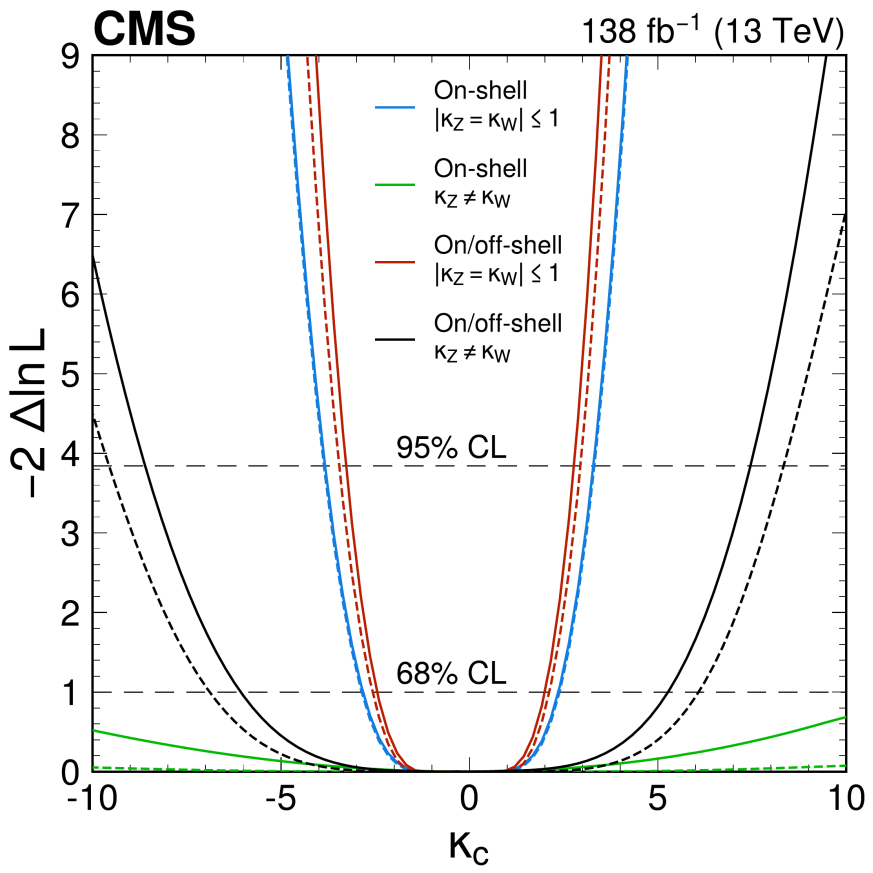
$$\sigma_{on} \propto \frac{g_p^2\, g_d^2}{\Gamma_H} \qquad \sigma_{off} \propto g_p^2\, g_d^2$$

$g_p$, $g_d$: production and decay couplings ($g_d \propto \kappa_Z$)

1. **Large $\kappa_c$** adds $c\bar{c}$ → H production and a larger width $\Gamma_H$
2. **On-shell** $\propto 1/\Gamma_H$: larger $\kappa_Z$ can hide larger $\Gamma_H$ ⇒ need $|\kappa_Z| \leq 1$
3. **Off-shell** has no $\Gamma_H$ ⇒ fixes $\kappa_Z$, assumption dropped

| $\kappa_c$ at 95% CL | observed | expected |
|---|---|---|
| On-shell, $\lvert\kappa_Z = \kappa_W\rvert \leq 1$ | [−3.7, 3.2] | [−3.8, 3.3] |
| NEW On + off-shell, $\lvert\kappa_V\rvert \leq 1$ | **[−3.3, 2.8]** | [−3.5, 2.9] |
| NEW On + off-shell, $\kappa_Z$, $\kappa_W$ free | [−8.6, 7.5] | [−9.6, 8.3] |

Higgs width: F. Errico (Tue) · $p_T^H$ spectrum → backup




On-shell (blue, green), on + off-shell (red, black); solid observed, dashed expected

***→ With $|\kappa_V| \leq 1$, H → 4ℓ matches the direct searches, under different assumptions***

# Where we stand

## CMS constraints on $\kappa_c$ at 95% CL

| Probe | Reference | observed | expected |
|---|---|---|---|
| $t\bar{t}H$ + $VH(H \to c\bar{c})$ | arXiv:2509.22535 | **$\lvert\kappa_c\rvert < 3.5$** | 2.7 |
| $t\bar{t}H(H \to c\bar{c})$ | arXiv:2509.22535 | $\lvert\kappa_c\rvert < 3.0$ | 3.3 |
| $VH(H \to c\bar{c})$ | arXiv:2205.05550 | $1.1 < \lvert\kappa_c\rvert < 5.5$ | 3.4 |
| On + off-shell $H \to 4\ell$, $\lvert\kappa_V\rvert \leq 1$ | arXiv:2607.23352 | **[−3.3, 2.8]** | [−3.5, 2.9] |
| On-shell $H \to 4\ell$, $\lvert\kappa_Z = \kappa_W\rvert \leq 1$ | arXiv:2502.05665 | [−3.7, 3.2] | [−3.8, 3.3] |
| On + off-shell $H \to 4\ell$, $\kappa_Z$, $\kappa_W$ free | arXiv:2607.23352 | [−8.6, 7.5] | [−9.6, 8.3] |
| cH ($\gamma\gamma$ + WW) | arXiv:2508.14988 | $\lvert\kappa_c\rvert < 47$ | 51 |
| $H \to J/\psi\,\gamma$ · sign-sensitive · J. Bierkens, Thu | arXiv:2411.15000 | [−166, 208] | [−128, 169] |

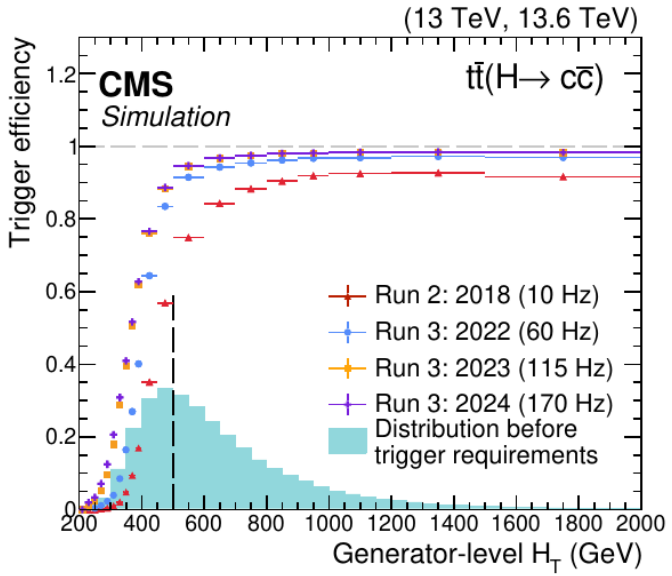
The results use different assumptions on the other Higgs couplings and are not directly comparable

***→ Direct searches and on + off-shell $H \to 4\ell$ both reach $\lvert\kappa_c\rvert \sim 3$, where flavour-compatible BSM lives***

# Toward Run 3

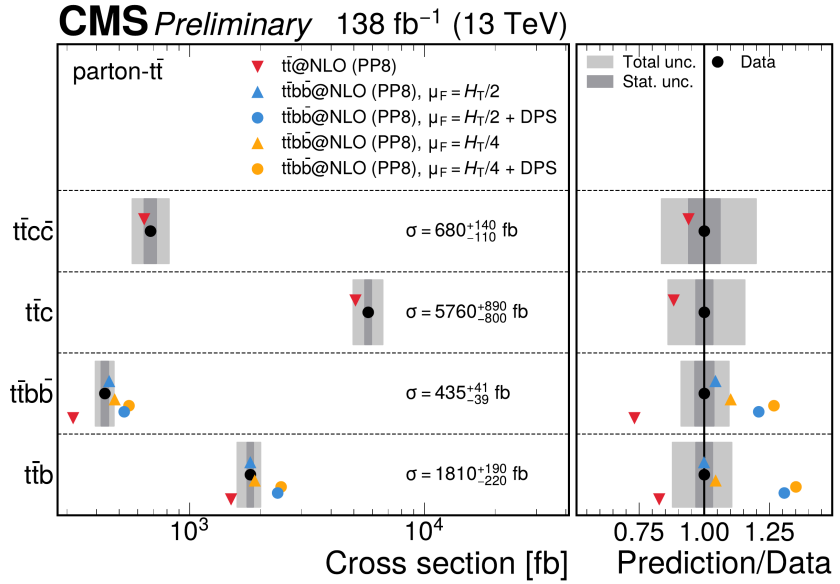
## More data, better tools, and the main $t\bar{t}H$ background measured

### Run 3: ML b/c tagging at the HLT




arXiv:2608.18331 · $t\bar{t}H(H \to c\bar{c})$ simulation: Run 2 (2018, 13 TeV) vs Run 3 (2022–24, 13.6 TeV) · J. Daniel (Mon)

### Run 2: $t\bar{t}c\bar{c}$ cross section measured




CMS-PAS-TOP-26-003 NEW, 138 fb$^{-1}$: $\sigma_{t\bar{t}c\bar{c}}$ = 680 ± 40 (stat) $^{+130}_{-100}$ (syst) fb

**More data**
direct searches are statistics-limited

**Better c tagging**
UParT (slide 3)

**Better triggers**
b/c tagging at the HLT (left)

**Better methods**
analysis and background modelling ($t\bar{t}c\bar{c}$, right)

***➜ Run 3: more data, plus better tagging, triggers and analysis methods***

# HL-LHC prospects

arXiv:2504.00672

## What the projection assumes, and why we want to beat it

### The projection (ATLAS + CMS combined)

- 14 TeV, **3000 fb$^{-1}$** per experiment (≈ 20× Run 2)
- **VH(H → $c\bar{c}$) only**: an ATLAS study, assumed to hold for CMS too
- Includes recent b and c tagging gains (CMS: UParT)

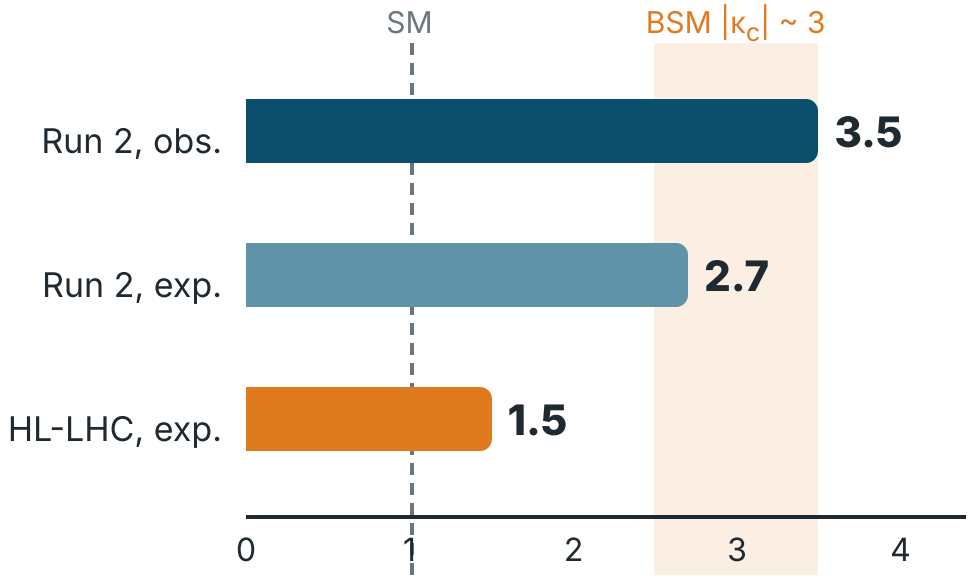
### Result

- **Expected $|\kappa_c| < 1.5$**, 30% better than the previous projection
- VH(H → $c\bar{c}$): ≈ 1.6σ, still no observation

**Our goal: beat it**

The report calls its projections **conservative targets**. Adding $t\bar{t}H$ and other channels, better taggers and analysis methods can push further

### Direct searches: allowed $|\kappa_c|$




Run 2: arXiv:2509.22535 · HL-LHC: arXiv:2504.00672
No bar excludes $\kappa_c = 0$ yet

$\kappa_c$ of order one: first sensitivity to whether the Higgs gives charm its mass

# Summary

## Does the Higgs give charm its mass?

The charm Yukawa is the most accessible second-generation quark coupling; CMS probes it **in the decay, in production and indirectly**

### Direct

$t\bar{t}H$ + VH($H \to c\bar{c}$)

**$|\kappa_c| < 3.5$** (2.7)

most stringent direct constraint, best expected sensitivity

### Production

H + charm, γγ + WW

**$|\kappa_c| < 47$** (51)

independent of the $H \to c\bar{c}$ decay

### Indirect

on + off-shell $H \to 4\ell$

**[−3.3, 2.8]**

with $|\kappa_V| \leq 1$ (about ±8 without)

### Next

Run 3 data

**better taggers and triggers**

$t\bar{t}c\bar{c}$ background already measured

Today **$|\kappa_c| \lesssim 3$–3.5**: entering BSM-motivated territory

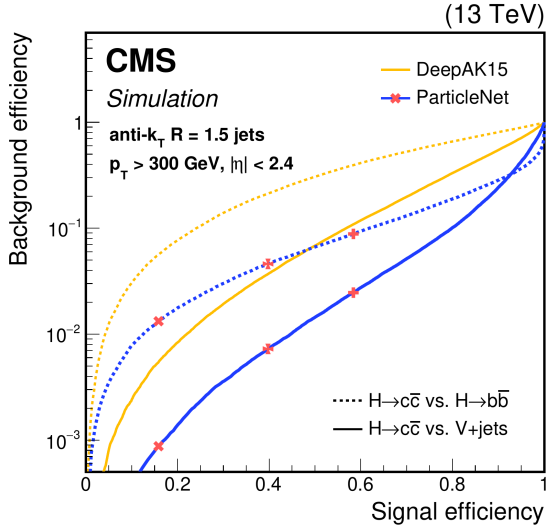
HL-LHC expected **$|\kappa_c| < 1.5$**: approaching, not yet reaching, sensitivity to $\kappa_c = 0$

***➜ Thank you!***

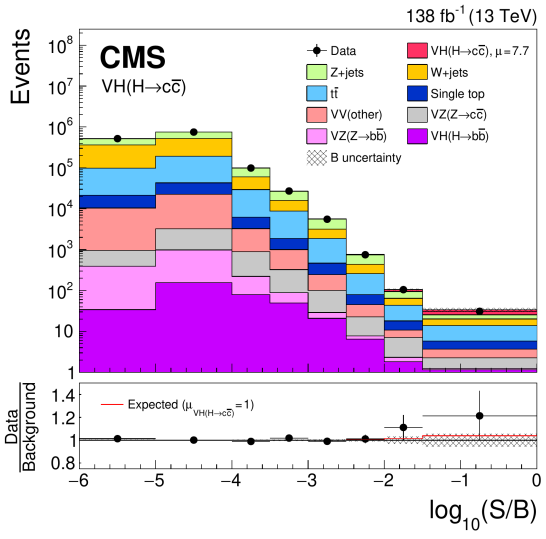
# Backup

# VH(H → $c\bar{c}$): categories and sensitivity

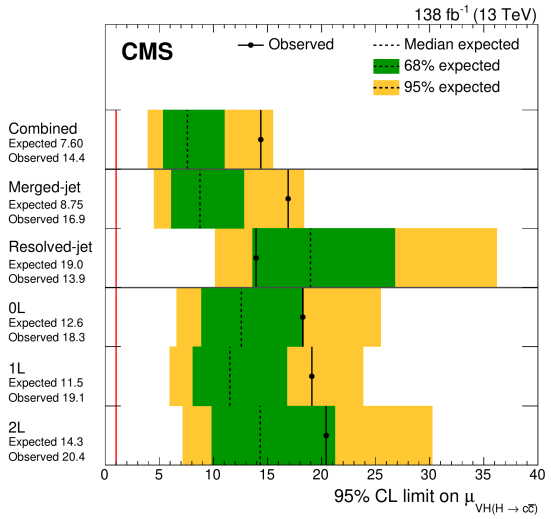
arXiv:2205.05550



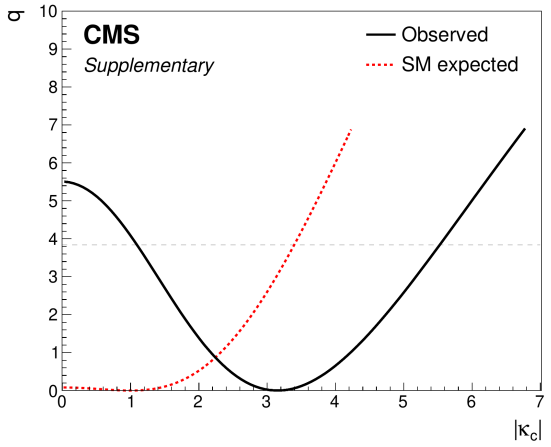

ParticleNet merged-jet H → $c\bar{c}$ tagging




Events vs $\log_{10}$(S/B)




μ limits per channel (inner band 68%)




|$\kappa_c$| scan (supplementary)

Best fit μ = 7.7 $^{+3.8}_{-3.5}$; merged μ = 8.7 $^{+4.6}_{-4.0}$, resolved μ = −9.5 ± 9.6 · $\sigma\mathcal{B}$ < 0.94 (0.50) pb · μ limits with CLs, $\kappa_c$ intervals from profile-likelihood scans

➜ *The upward fluctuation comes from the merged-jet category*

# VH(H → $c\bar{c}$): analysis details

arXiv:2205.05550 · strategy as in arXiv:1912.01662

**Topologies** Two, orthogonal: H candidate from a large-R (R = 1.5) jet with $p_T$ above (merged) or below (resolved) 300 GeV; same lepton channels

**Resolved** DeepJet CvsL/CvsB; the two highest-CvsL jets form the H candidate; leading-jet cuts at ≈40% c efficiency (≈4% light, ≈16% b mistag). c-jet energy regression, FSR-jet recovery and a 2ℓ kinematic fit improve the $m(H_{cand})$ resolution by up to 20%. BDT per channel, fit to the BDT score

**Merged** ParticleNet $c\bar{c}$ tagger, 4–7× better rejection than DeepAK15; 3 working points (≈58, 40, 16% $c\bar{c}$ efficiency) define 3 purity categories. ParticleNet mass regression: ≈50% better mass resolution. BDTs decorrelated from $m(H_{cand})$; fit to $m(H_{cand})$

**Backgrounds** V+jets and $t\bar{t}$ normalised in dedicated control regions, following the 2016 analysis (V+jets split by jet flavour, free rate parameters)

**VZ validation** Same analysis, resolved BDTs retrained with VZ as signal, merged unchanged. $\mu_{VZ} = 1.01\ ^{+0.23}_{-0.21}$; 4.4σ merged, 3.1σ resolved, 5.7σ combined (exp. 4.7, 3.3, 5.9)

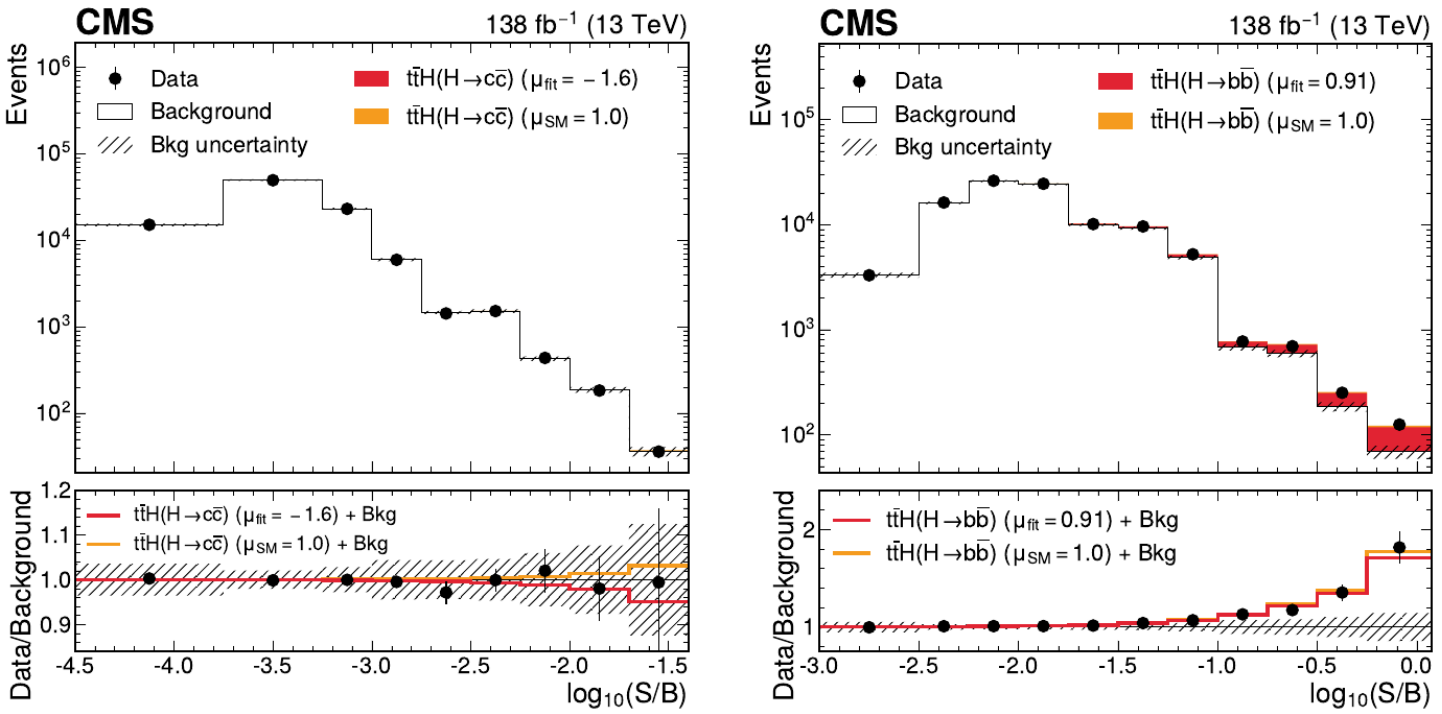
| Source | $\Delta\mu/\Delta\mu_{tot}$ |
|---|---|
| **Statistical** | **85%** |
| Bkg. normalisations | 37% |
| Experimental | 48% |
| Simulated sample sizes | 37% |
| c-jet identification | 23% |
| Jet energy scale, resolution | 15% |
| Simulation modelling | 11% |
| Luminosity | 6% |
| Lepton identification | 4% |
| Theory | 22% |
| Backgrounds | 17% |
| Signal | 15% |

For best-fit $\mu = 7.7\ ^{+3.8}_{-3.5}$; simulation size driven by negative weights in NLO V+jets

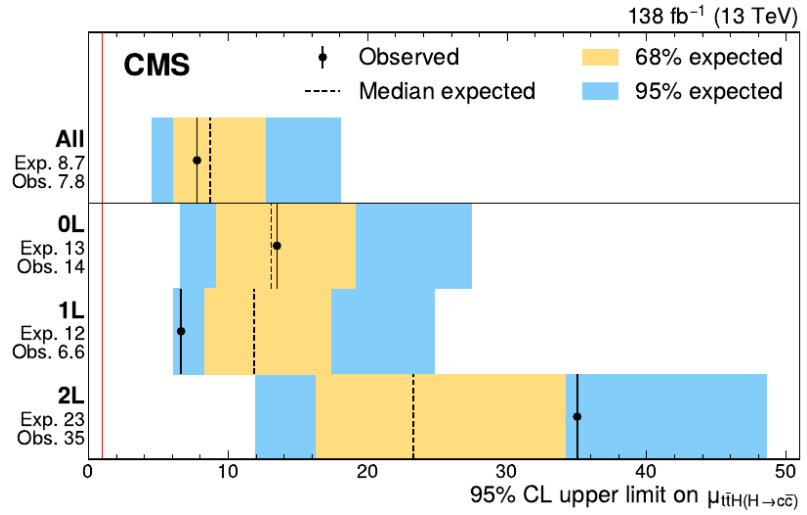
***➜ Statistics-limited; c tagging and simulation size come next***

# $t\bar{t}H(H \to c\bar{c})$: event yields

arXiv:2509.22535




Event yields vs $\log_{10}$(S/B); the $t\bar{t}H(c\bar{c})$ panel shows the published best fit




$\mu_{t\bar{t}H(c\bar{c})}$ limits per channel (inner band 68%)

**→ $t\bar{t}H(c\bar{c})$ best-fit $\mu = -1.6 \pm 4.5$: compatible with zero and with the SM**

# $t\bar{t}H(H \to c\bar{c})$: flavour tagging and by-products

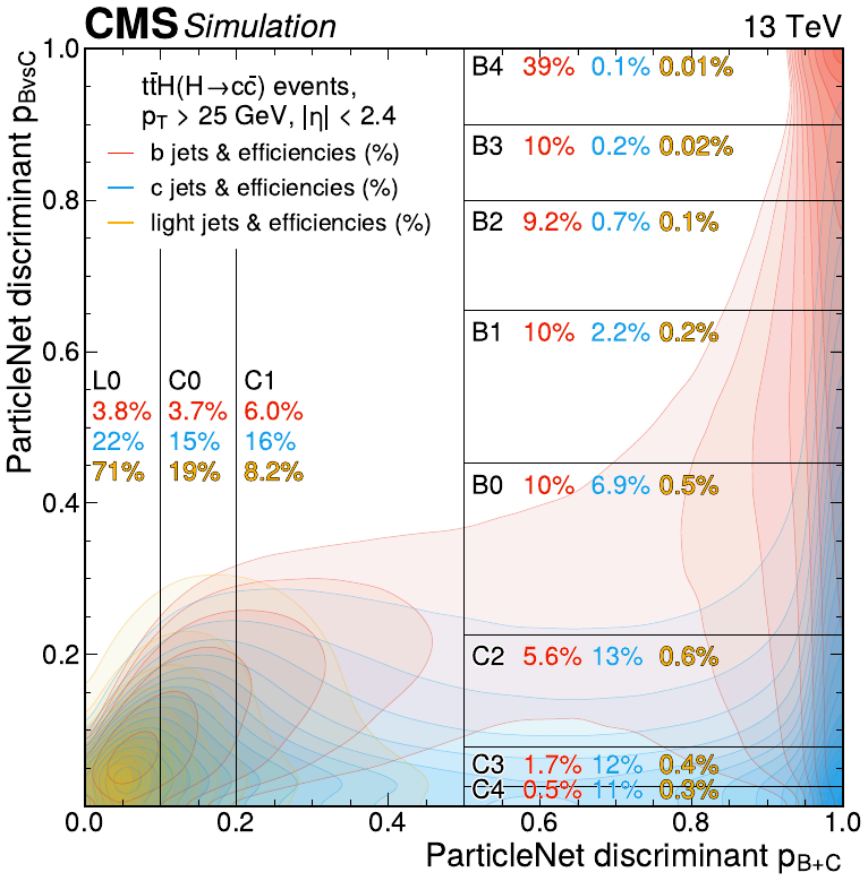
arXiv:2509.22535

## Tagging

- Each jet gets ParticleNet scores; the (b+c vs light, b vs c) plane is split into 2D categories
- ParT event classifier separates $t\bar{t}H(c\bar{c})$, $t\bar{t}H(b\bar{b})$, $t\bar{t}Z$, $t\bar{t}$ +HF

## By-products

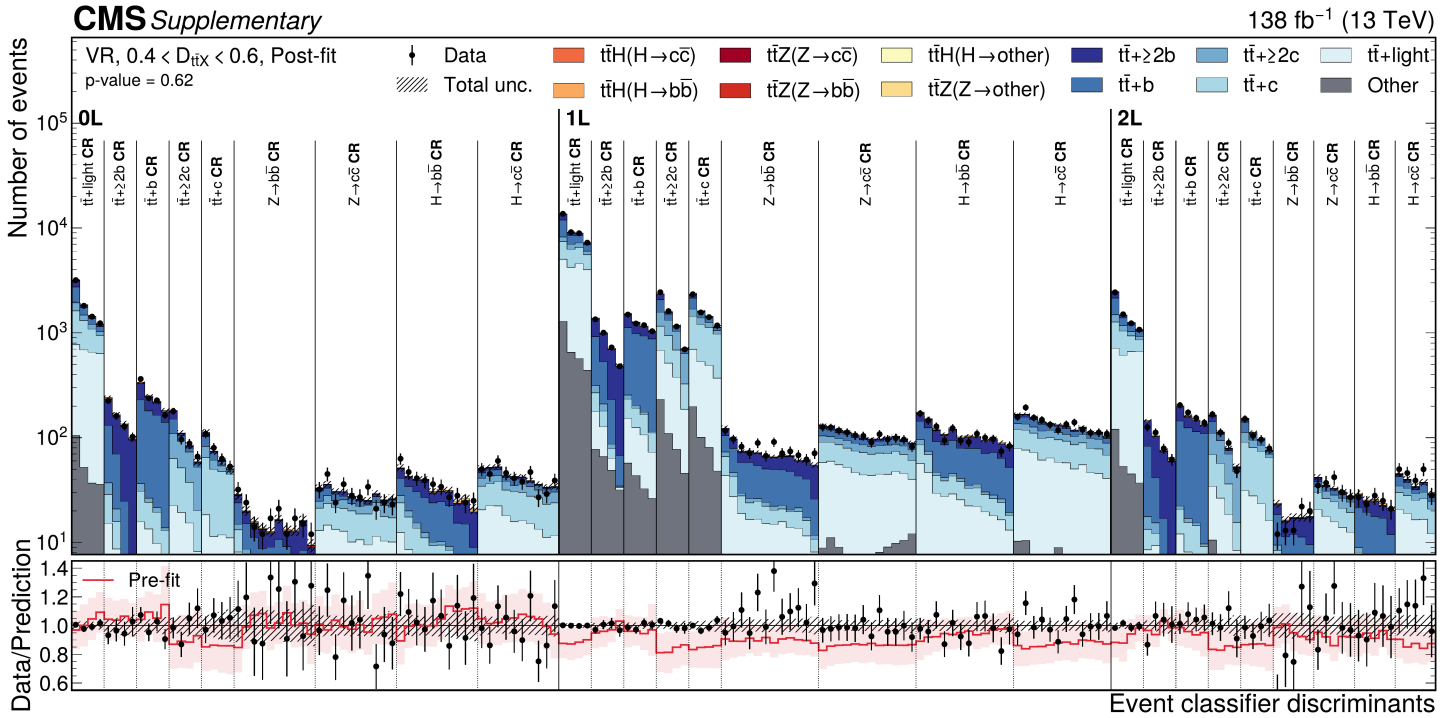
- $t\bar{t}Z(Z \to c\bar{c})$ standard candle: $\mu = 1.0 \pm 0.8$
- $t\bar{t}H(H \to b\bar{b})$: $\mu = 0.91\ ^{+0.26}_{-0.22}$, $4.4\sigma$ ($4.5\sigma$ exp.)




b, c, light jets in the ParticleNet plane and the 2D tagging categories

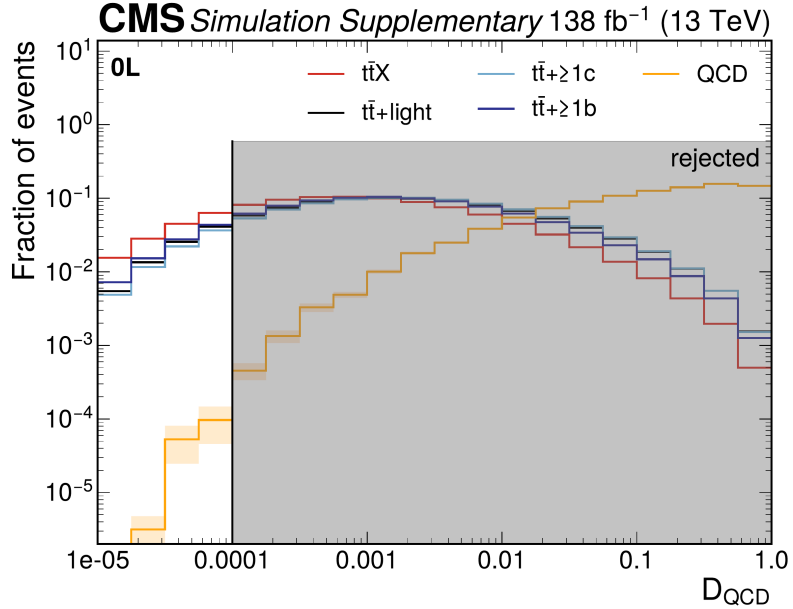
# $t\bar{t}H$: validation region and QCD rejection

arXiv:2509.22535 (supplementary)




Validation region 0.4 < $D_{t\bar{t}X}$ < 0.6, same SR/CR structure (CR-like 0.40–0.58, SR-like 0.58–0.60): good agreement

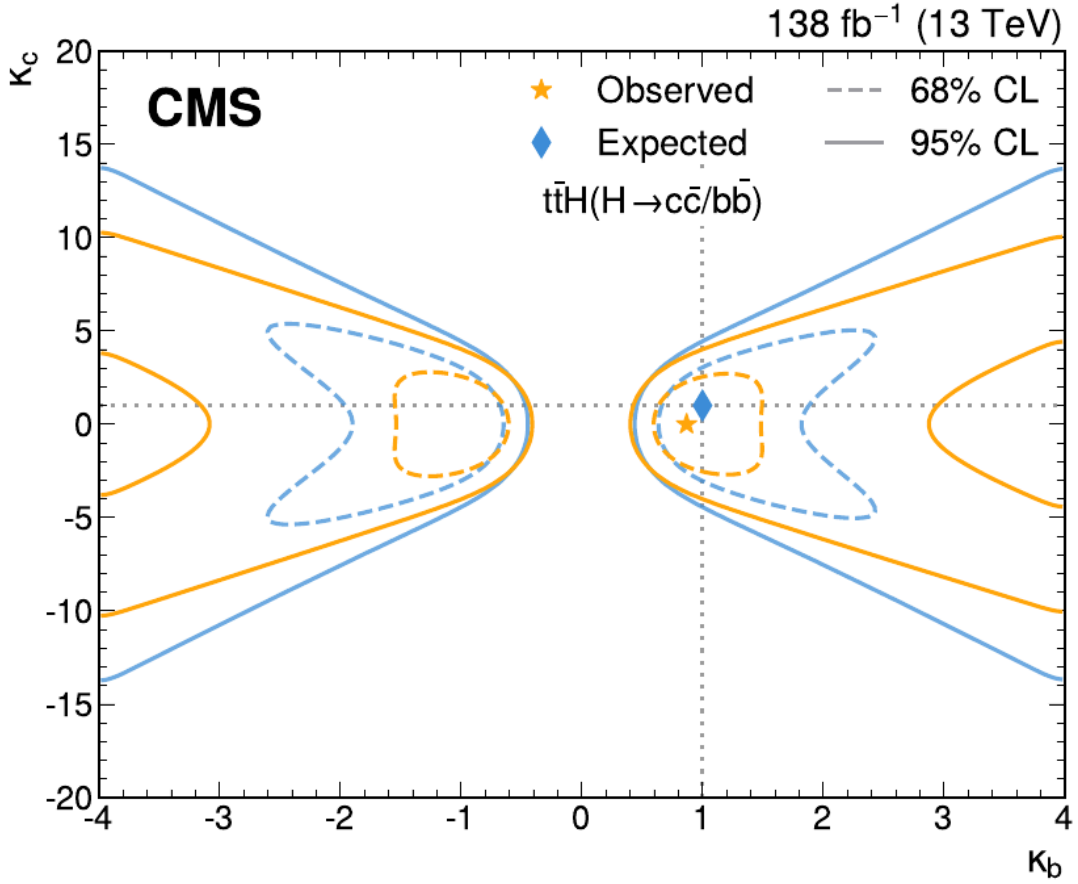
- c-enriched regions need ≥2 c jets ($t\bar{t}$+c: ≥1); CR choice weights 100, 12, 4, 2, 1 on the $t\bar{t}$+light, +c, +≥2c, +b, +≥2b scores
- $t\bar{t}$+b, +≥2b from a dedicated 4FS $t\bar{t}b\bar{b}$ sample




0ℓ only: $D_{QCD}$ < $10^{-4}$ (grey area rejected) suppresses QCD by nearly four orders of magnitude, to <1% of $t\bar{t}$+jets; QCD simulation only trains the classifier

# $t\bar{t}H$: the ($\kappa_b$, $\kappa_c$) plane

arXiv:2509.22535




orange: observed, blue: expected; dashed 68%, solid 95% CL

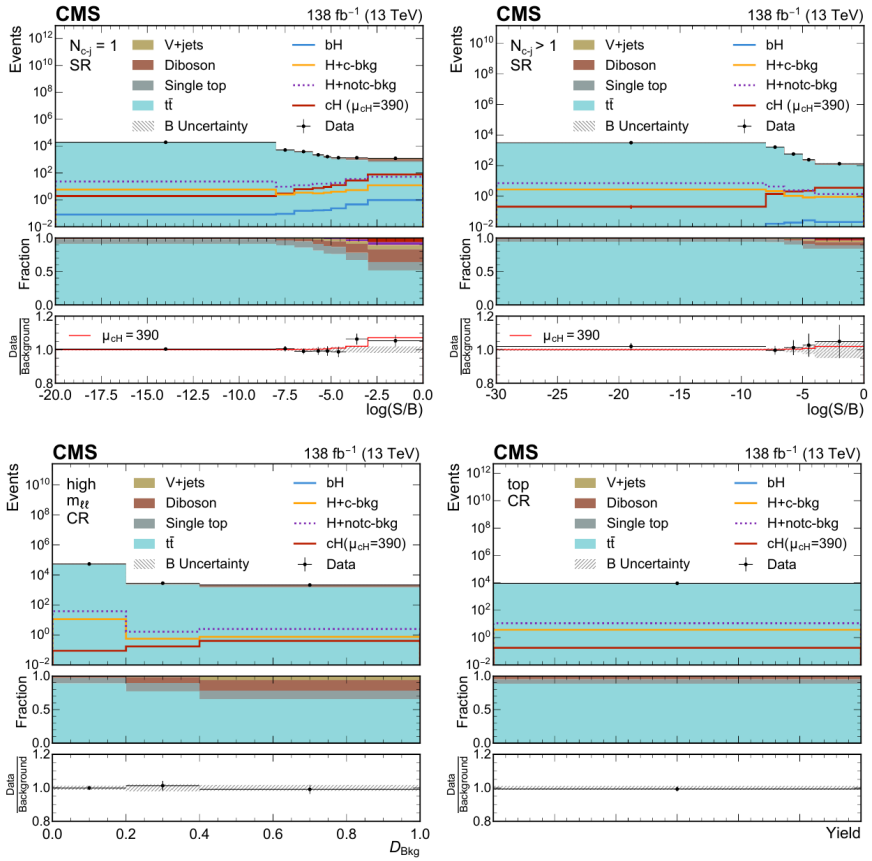
The simultaneous $t\bar{t}H(H \to b\bar{b})$ + $t\bar{t}H(H \to c\bar{c})$ fit constrains both Yukawas

Signal strengths: $t\bar{t}H(b\bar{b})$ $0.91\ ^{+0.26}_{-0.22}$ · $t\bar{t}H(c\bar{c})$ $-1.6 \pm 4.5$ · $t\bar{t}Z(b\bar{b})$ $1.47\ ^{+0.45}_{-0.41}$ · $t\bar{t}Z(c\bar{c})$ $1.02\ ^{+0.79}_{-0.84}$

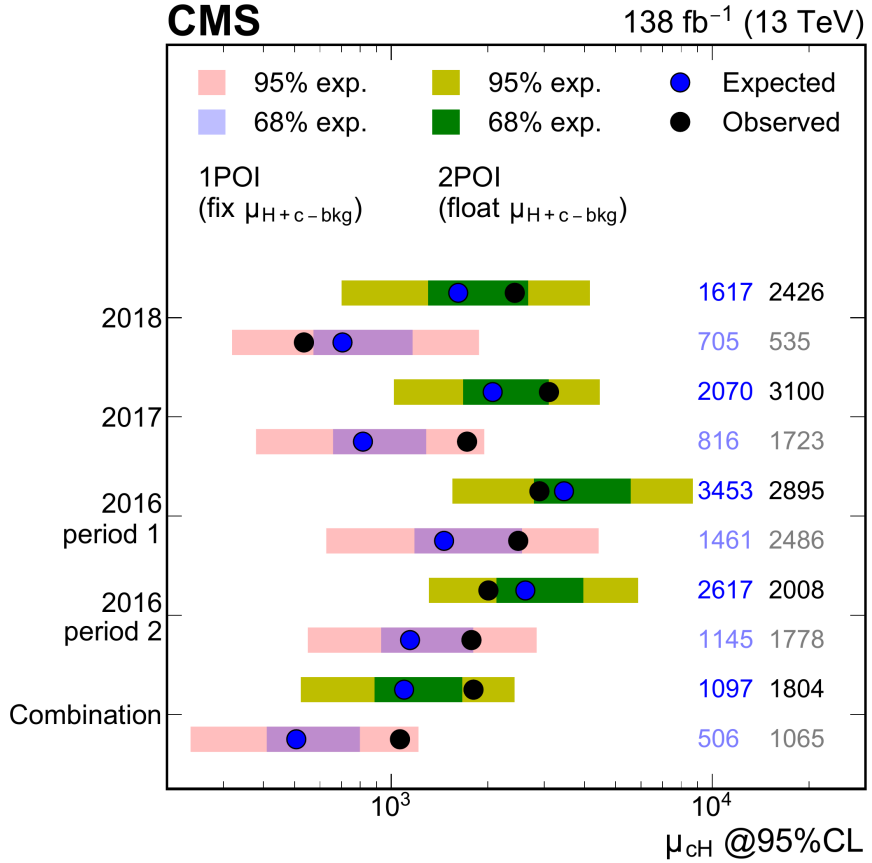
**$|\kappa_c| < 4.2$ obs. with $\kappa_b$ fitted**

# cH(H → WW): post-fit and per-period limits

arXiv:2508.14988




1-POI post-fit BDT discriminants




$\mu_{cH}$ limits per data-taking period, 1 and 2 POI

➜ *Best fit (1 POI): $\mu_{cH} = 390\ ^{+380}_{-270}$, within 1.5σ of the SM; statistics-limited, then flavour-scheme and ggH heavy-flavour modelling*

# cH(H → WW): floating the H+c background

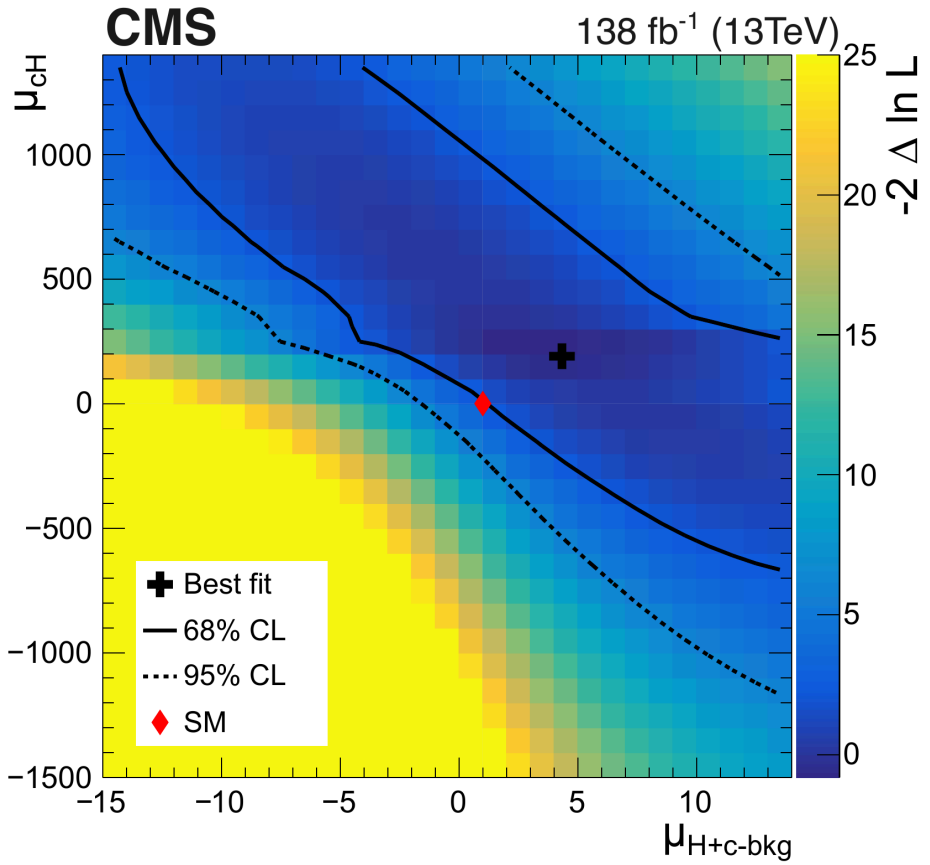
arXiv:2508.14988

**1 POI** only $\mu_{cH}$ floats; all other Higgs processes (ggH, VBF, VH, $t\bar{t}H$, bH) are fixed to their SM prediction, with a 50% uncertainty on their heavy-flavour part

**2 POI** the charm-associated part, $\mu_{H+c\ bkg}$, also floats; correlation with cH: −87%

2-POI limit: $\mu_{cH} < 1804$ (1097)

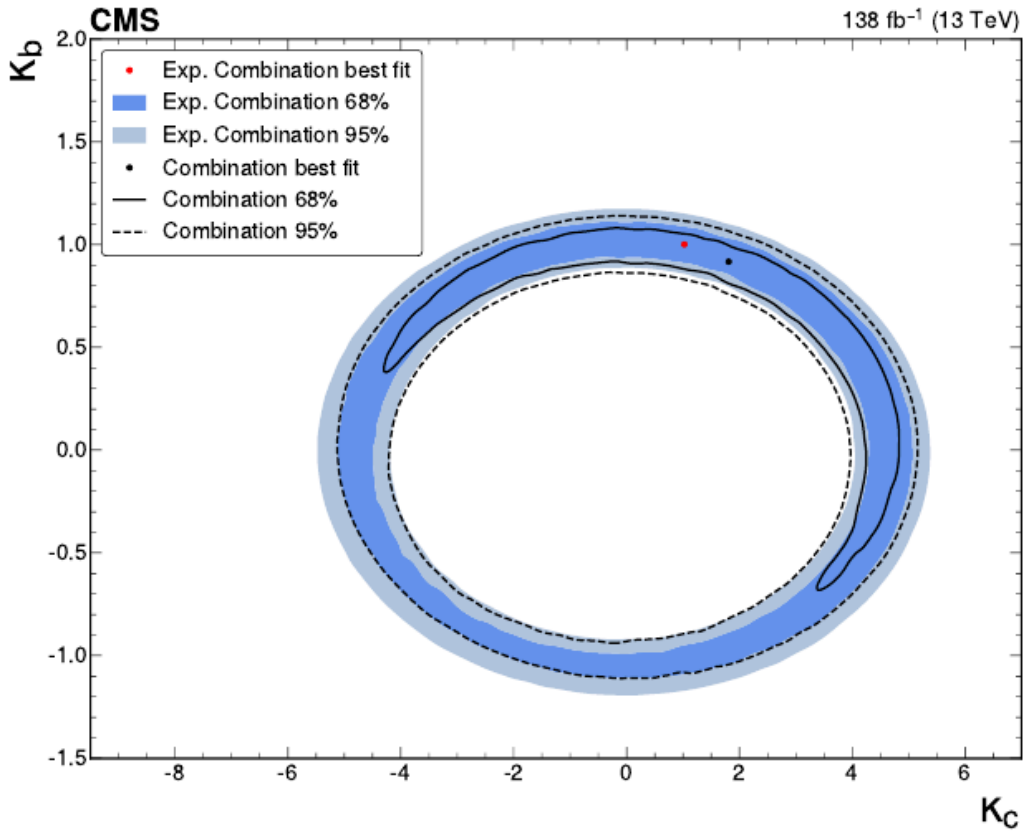
The combination with γγ uses the 1-POI fit



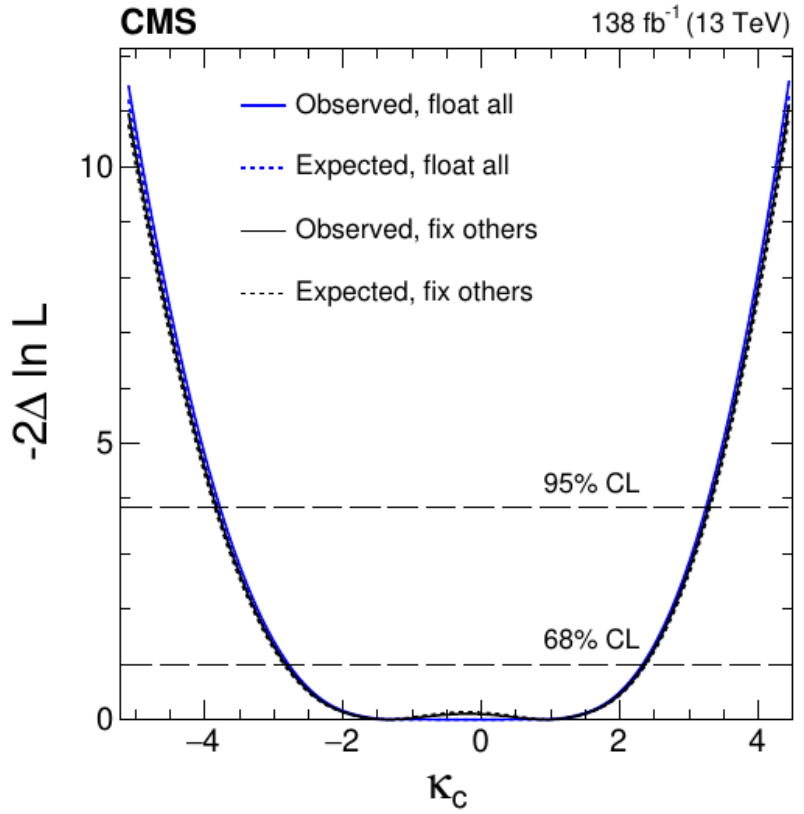

WW: $\mu_{cH}$ vs $\mu_{H+c\ bkg}$

***➜ Floating the H+c background weakens the WW limit from 1065 to 1804***

# Higgs $p_T$ spectrum and on-shell H → 4ℓ




arXiv:2504.13081: ($\kappa_b$, $\kappa_c$) from the Run 2 $p_T^H$ combination (γγ, ZZ, resolved ττ), $\mathscr{B}$ coupling-dependent; 2D contours only




arXiv:2502.05665 (γH + on-shell H → 4ℓ): $\kappa_c$ scan; solid obs., dashed exp.

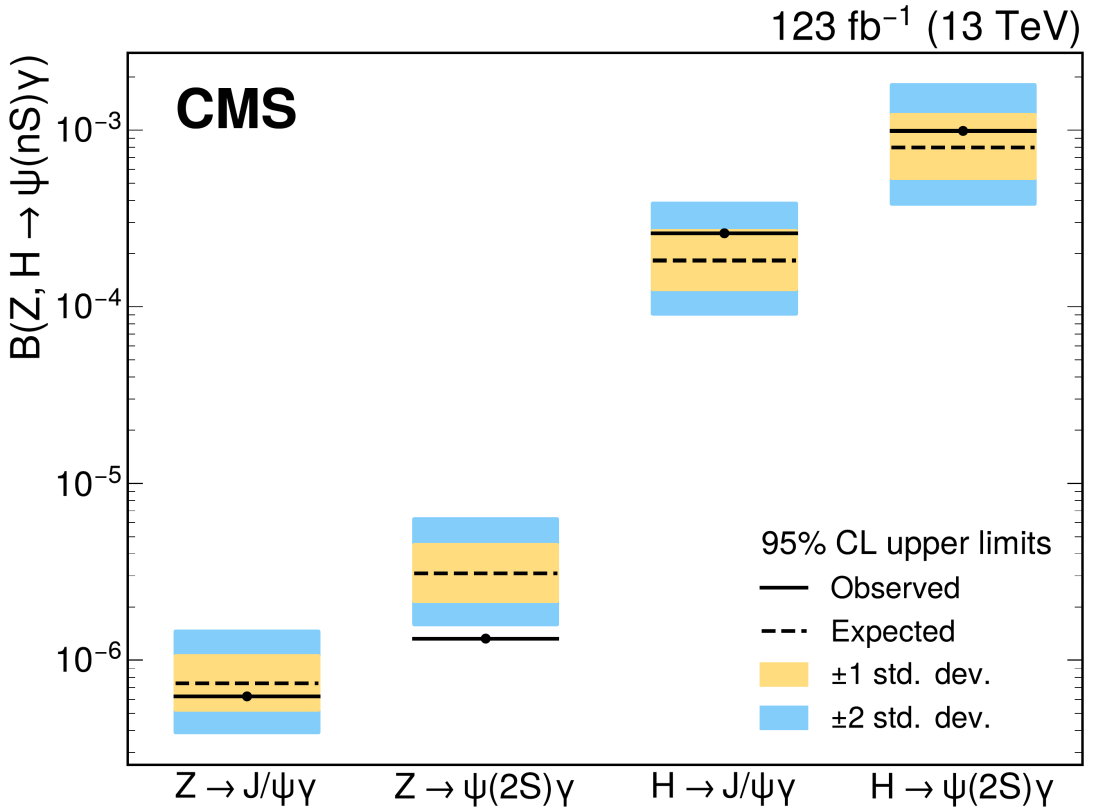
# Rare exclusive decays: H → J/ψ γ, ψ(2S) γ

arXiv:2411.15000

Direct ($\propto \kappa_c$) and indirect (H → γγ* → γ J/ψ) amplitudes interfere ⇒ **sign sensitivity**

123 fb$^{-1}$: $\mathcal{B}$(H → J/ψ γ) < 2.6 × 10$^{-4}$, $\mathcal{B}$(H → ψ(2S) γ) < 9.9 × 10$^{-4}$

**$\kappa_c/\kappa_\gamma \in (-157, 199)$**
**with $\kappa_\gamma = 1$: $\kappa_c \in (-166, 208)$**
expected (−128, 169) · 95% CL




95% CL upper limits on $\mathcal{B}$(Z, H → ψ(nS) γ), observed and expected

# H → μμ: a second-generation Yukawa

arXiv:2009.04363

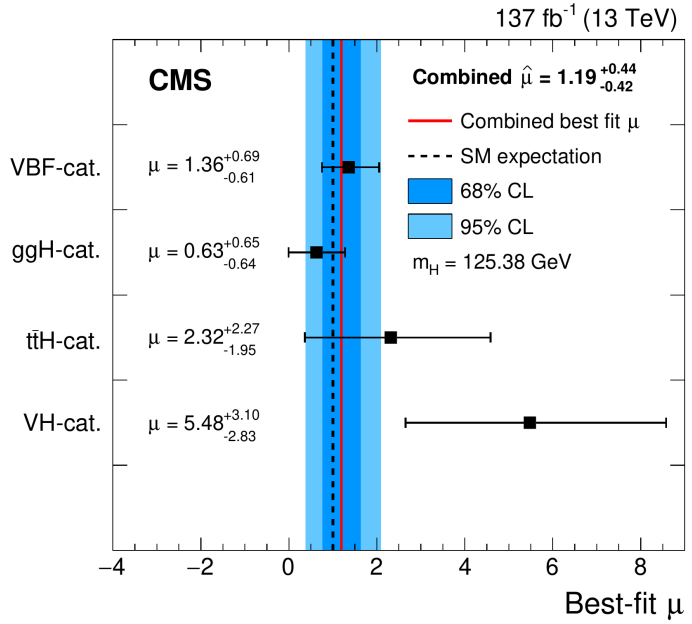
137 fb$^{-1}$, categories ggH, VBF, VH, $t\bar{t}H$

Excess of **3.0σ** (2.5σ exp.): first evidence for H decays to second-generation fermions

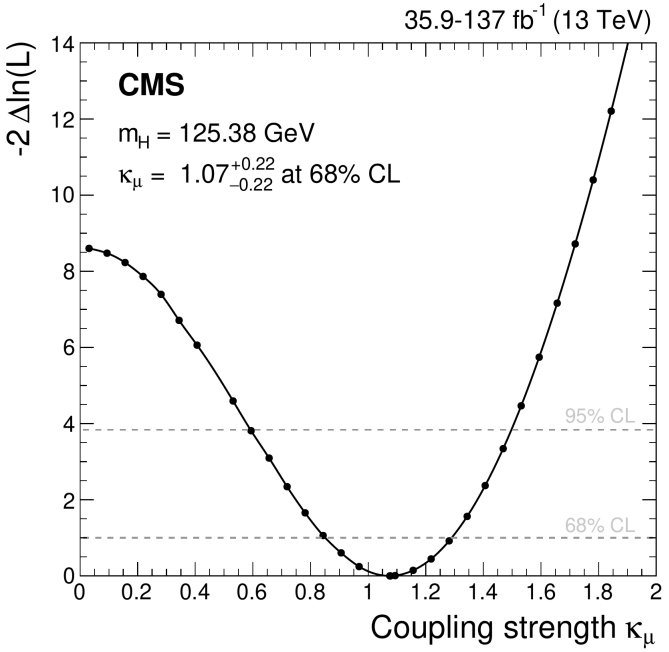
$\mu = 1.19\ ^{+0.40}_{-0.39}\ (\text{stat})\ ^{+0.15}_{-0.14}\ (\text{syst})$

$\kappa_\mu = 1.07$; 95% CL: $0.59 < \kappa_\mu < 1.50$ (6 free κ, SM total width)

Rare decays incl. μμ: L. D'Eramo's plenary (Wed)




Best-fit μ per production category and combined

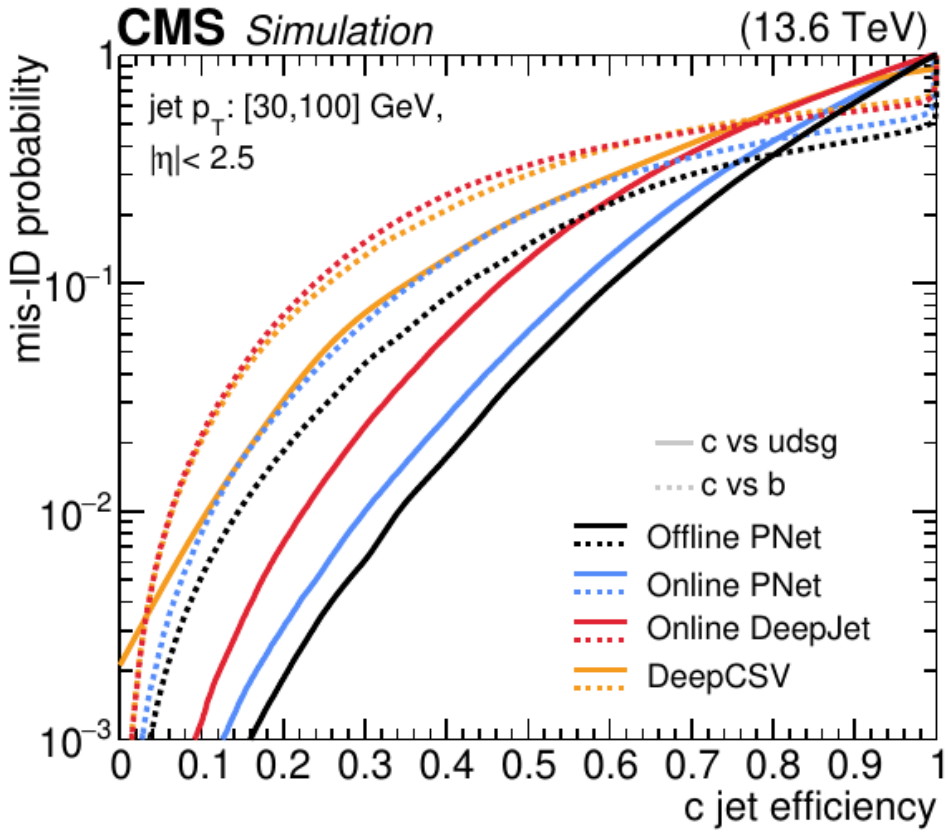



$-2\Delta\ln L$ vs $\kappa_\mu$

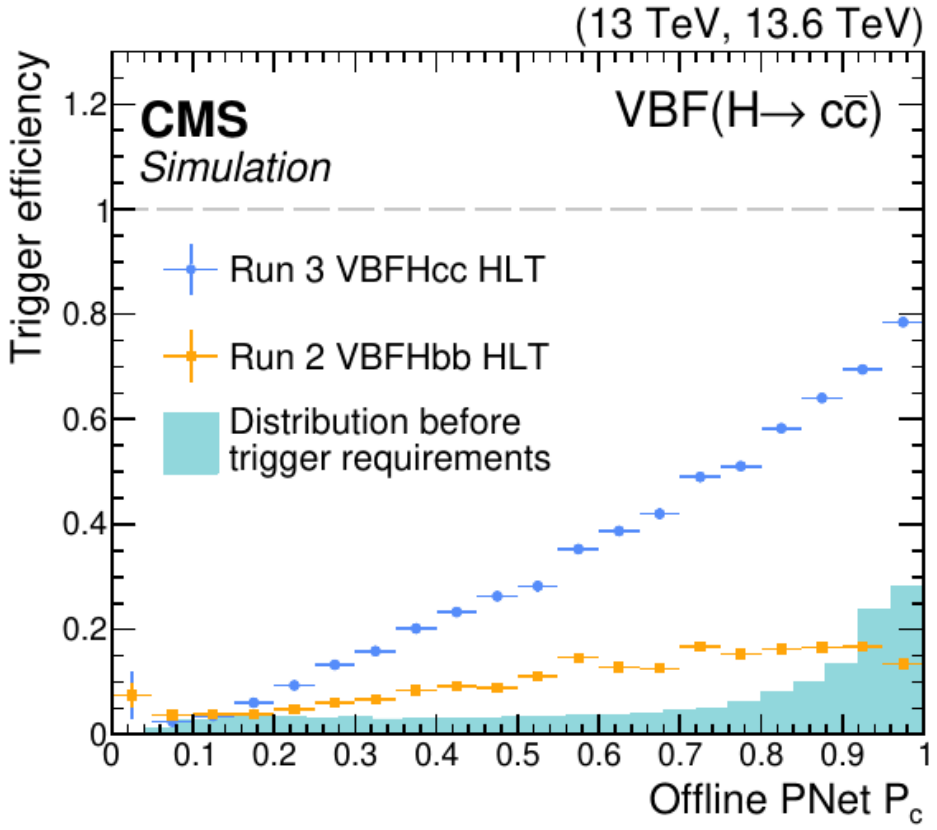
***➜ The muon Yukawa is SM-like within about 20% (68% CL); charm is the next test***

# Run 3 HLT flavour tagging

arXiv:2608.18331




Simulation, jet $p_T$ 30–100 GeV: light (solid) and b (dashed) misidentification vs c-jet efficiency, online vs offline taggers




Simulation: VBF H → cc̄ trigger efficiency vs offline ParticleNet c score, Run 3 VBF H → cc̄ vs Run 2 VBF H → bb̄ trigger

# Likely questions

- **How large is the VH excess?** Best fit $\mu_{VH} = 7.7\ ^{+3.8}_{-3.5}$, about 1.9σ above μ = 1 (estimated from the quoted uncertainty; the paper calls it a small upward fluctuation). This is why VH alone excludes $|\kappa_c| < 1.1$ and why $t\bar{t}H$ alone (3.0) beats the combination (3.5) on observed
- **What limits the direct searches?** Statistics first. $t\bar{t}H(c\bar{c})$: statistical 74% of the total uncertainty, then $t\bar{t}$+c and $t\bar{t}$+light modelling (≈32%, ≈29%). $VH(c\bar{c})$: simulation sample size (≈37%), c-tagging efficiency (≈23%), theory (≈22%)
- **And ATLAS?** VBF + VH(H → $c\bar{c}$) (arXiv:2511.21911): $|\kappa_c| < 4.7$ (3.9 exp.), same assumptions as CMS (other couplings SM, no BSM decays); its VBF part includes 51.5 $fb^{-1}$ of Run 3 data
- **Why is cH an order of magnitude weaker than H → $c\bar{c}$?** cH $\sigma\mathscr{B}$ is tiny (≈0.2 fb in γγ) and ggH produces the same H+c final state independently of $\kappa_c$; in H → $c\bar{c}$, $\kappa_c$ changes the branching fraction of Higgs bosons from much larger production modes (VH, $t\bar{t}H$)
- **Why is cH(γγ) obs. (38) much better than exp. (73)?** The best fit is negative, $\mu_{cH} = -230\ ^{+170}_{-251}$
- **Why no $\kappa_c$ from boosted H → $c\bar{c}$?** The paper quotes only μ; the result is ggH-dominated
- **How much of the Run 3 trigger gain is the tagger?** Part of it is rate budget: 10 Hz in 2018 vs 60, 115, 170 Hz in 2022, 2023, 2024
- **Are the H → 4ℓ intervals sign-sensitive?** Not really: the small asymmetry comes from charm-loop interference in ggH and H → gg; the constraint is essentially on $\kappa_c^2$

# References


- VH(H → $c\bar{c}$): arXiv:2205.05550
- $t\bar{t}$H(H → $c\bar{c}$) + VH combination: arXiv:2509.22535
- VH(H → $c\bar{c}$) with 2016 data (analysis strategy): arXiv:1912.01662
- Boosted H → $c\bar{c}$: arXiv:2211.14181
- cH(H → γγ): arXiv:2503.08797
- cH(H → WW) and combination: arXiv:2508.14988
- Differential combination Run 2: arXiv:2504.13081
- γH + H → 4ℓ light Yukawas: arXiv:2502.05665
- Off-shell + on-shell H → 4ℓ: arXiv:2607.23352
- H → J/ψ γ: arXiv:2411.15000
- H → μμ: arXiv:2009.04363
- Jet flavour tagging in Run 3: CMS-DP-2025-081 · DeepJet: arXiv:2008.10519 · ParT: arXiv:2202.03772
- HLT flavour tagging: arXiv:2608.18331
- $t\bar{t}$+b/c cross sections: CMS-PAS-TOP-26-003
- ATLAS VBF + VH(H → $c\bar{c}$): arXiv:2511.21911
- HL-LHC: arXiv:2504.00672
- Higgs combination: arXiv:2602.18611
- Spontaneous flavour violation: arXiv:1811.00017 · $\kappa_c$ in SFV ("How charming can the Higgs be?"): arXiv:2410.05236
- SM Higgs branching ratios: arXiv:1610.07922
- HL-LHC Yellow Report: arXiv:1902.00134